AnchorSum: Improving Multi-Document Summarization Faithfulness through NLI and Entity-Gated Iterative Refinement

Anonymous ACL submission

Abstract

Abstractive multi-document summarization systems, which are trained to maximize conditional likelihood, frequently exhibit hallucination by generating entities, events, and statistics that are not supported by any source document. We introduce **AnchorSum**, a modular, training-free pipeline designed to address this issue through three closely integrated mechanisms: *anchor-conditioned draft generation*, which grounds the generation process in high-salience named entity constraints extracted from the source corpus; *dual-mode faithfulness auditing*, which independently verifies sentence-level entailment using a cross-encoder NLI model and entity-level grounding through a conjunctive hallucination filter; and *flag-guided iterative revision*, which re-prompts the generator with structured corrective feedback from external neural verifiers, rather than relying on a potentially biased self-assessment. When evaluated on 500 Multi-News instances (Fabbri et al., 2019) across six automatic metrics and on 50 instances via a dual-judge LLM-as-judge panel (Liu et al., 2023; Verga et al., 2024), AnchorSum achieves a 6.3% relative improvement in SummaCConv over the unaugmented generator and surpasses all baselines—including PRIMERA-MultiNews and PEGASUS-multi_news—across every dimension of faithfulness. A controlled study of revision depth reveals a novel *verifier exploitation* failure mode: a second revision cycle increases SummaCConv by $+0.185$ while reducing BARTScore $s \rightarrow d$ by -2.566 nats, indicating that NLI-driven iterative refinement can optimize the auditor rather than the summary itself.

1 Introduction

Multi-document summarization (MDS) necessitates a model capable of synthesizing information from multiple heterogeneous sources into a coherent, comprehensive, and factually grounded summary. This task imposes simultaneous demands on coverage, consistency, and faithfulness, which no single optimization objective currently satisfies. Contemporary neural approaches, including fine-tuned sequence-to-sequence models (Lewis et al., 2020; Zhang et al., 2020a) and instruction-tuned large language models (Dubey et al., 2024), optimize conditional log-likelihood over reference text. However, likelihood maximization is an inadequate surrogate for faithfulness. Maynez et al. (2020) demonstrate that a significant fraction of generated summary sentences are not entailed by source documents, even for models with strong automatic metric performance. This discrepancy between fluency and factual grounding constitutes the central unresolved challenge in abstractive summarization.

Problem Statement

Formally, given a source corpus $\mathcal{D}{=}d_1 \| \cdots \| d_K$ of K documents, a summarization model parameterized by $\boldsymbol{\theta}$ produces $\hat{S}{=}\arg\max_S P(S \mid \mathcal{D}; \boldsymbol{\theta})$. This objective admits two distinct hallucination failure modes (Ji et al., 2023): *intrinsic* hallucinations, where the summary contradicts information present in $\mathcal{D}$, and *extrinsic* hallucinations, where the summary introduces entities or facts absent from $\mathcal{D}$ entirely. Both arise because the model's parametric memory and the fluency bias of likelihood training jointly allow high-confidence generation of content that is plausible but unsourced. Existing post-hoc correction approaches either rely on the same LLM to self-assess its own output (Madaan et al., 2023)—a setup prone to sycophantic reinforcement of existing errors—or apply faithfulness metrics as soft training signals, requiring costly re-training that limits applicability to new architectures (Gehrmann et al., 2023). A training-free, modular solution that intercepts both failure modes via independent, externally grounded verification signals remains an open challenge.

Contributions

We introduce **AnchorSum**, a four-phase pipeline that addresses the faithfulness problem without any task-specific fine-tuning. Our principal contributions are:

- **Anchor-conditioned generation.** A RoBERTa-base named-entity pipeline (Liu et al., 2019; Honnibal et al., 2020) extracts a corpus-frequency-ranked anchor set $\mathcal{A}$ that serves as both a generative soft constraint and a post-hoc verification target, grounding the draft in the most salient entities of the source cluster (Huang et al., 2021; Duan et al., 2024).

- **Dual-mode faithfulness auditing.** Two complementary modules—a cross-encoder NLI auditor (DeBERTa-V3-Large; He et al., 2023; Laban et al., 2022) and a conjunctive entity-hallucination filter—address non-overlapping failure modes in Maynez et al.'s hallucination taxonomy: sentence-level semantic inconsistency and extrinsic entity introduction respectively.

- **Externally grounded iterative revision.** Flag-guided revision supplies the generator with structured corrective feedback from *external* neural verifiers, explicitly avoiding sycophantic self-evaluation (Madaan et al., 2023) while preserving fluency and discourse coherence (Tang et al., 2024).

- **Verifier exploitation as a novel failure mode.** We characterize and empirically document *verifier exploitation*— a pathological regime in which additional revision cycles optimize the NLI auditor rather than the underlying summary quality—and provide a principled metric-triangulation methodology for detecting it (Yuan et al., 2021; Zha et al., 2023).

The remainder of the paper is organized as follows. §2 reviews related work. §3 formalizes AnchorSum's four phases. §4 presents ablation studies over 500 Multi-News instances. §5 reports LLM-as-judge comparative evaluation. §6 provides cross-method convergence analysis. Appendices A–C supply evaluation protocols, full judge reports, and detailed component architecture diagrams.

2 Related Work

Multi-document summarization. In the domain of Neural MDS, fine-tuned encoder-decoder architectures have predominantly been utilized. Notably, BART-large-CNN (Lewis et al., 2020) and PEGASUS-multi_news (Zhang et al., 2020a) were pre-trained on single-document corpora or had small token window, which results in suboptimal transfer to multi-source contexts. This limitation is highlighted by the Multi-News benchmark (Fabbri et al., 2019), which reveals significantly lower ROUGE scores compared to models specifically designed for this purpose. PRIMERA-MultiNews (Xiao et al., 2022) addresses this issue by incorporating entity-pyramid masking into the LED backbone, thereby explicitly enhancing multi-source synthesis and achieving state-of-the-art ROUGE scores. However, it remains susceptible to the same faithfulness issues that broadly affect autoregressive objectives. AnchorSum serves as a complement rather than a replacement for fine-tuned backbones; it is architecture-agnostic and can be applied during inference without altering model weights, thus allowing it to be integrated with any sufficiently capable generator.

Faithfulness in abstractive summarization. Maynez et al. (2020) conducted a seminal large-scale annotation study, establishing that a significant proportion of abstractive summary sentences are unfaithful, and distinguishing between intrinsic and extrinsic hallucinations. Pagnoni et al. (2021) further decomposed faithfulness errors into a detailed typology, including predicate, entity, circumstance, and link errors, thereby motivating the development of specialized detection strategies. Ji et al. (2023) surveyed the broader hallucination landscape across natural language generation tasks, demonstrating that entity-level grounding failures are both widespread and resistant to training signals based solely on fluency. AnchorSum directly operationalizes this taxonomy: the entity guard targets extrinsic hallucinations at the span level, while the NLI module addresses intrinsic semantic inconsistencies at the sentence level.

NLI-based faithfulness detection and correction. The most relevant prior research employs NLI inference for faithfulness scoring. Laban et al. (2022) demonstrate that sentence-level NLI scoring—aggregated through convolution over source sentences—achieves a strong correlation with human judgments of faithfulness and circumvents the

distributional mismatch associated with document-level NLI inference. Zha et al. (2023) develop a more comprehensive alignment scoring function that spans seven NLU tasks, offering a cross-framework faithfulness signal that is not vulnerable to the NLI-specific distributional artifacts exploited by verifier gaming. Yuan et al. (2021) adapt a pre-trained generative model as a conditional log-probability evaluator, resulting in a metric sensitive to register and discourse coherence rather than solely factual content. AnchorSum synthesizes insights from all three approaches: SummaCConv-style sentence auditing initiates revision flags, while AlignScore and BARTScore collectively function as cross-framework diagnostics for verifier exploitation.

Iterative refinement and self-correction. Madaan et al. (2023) illustrate that LLMs can iteratively enhance their outputs through self-generated feedback, yet they document a sycophancy failure mode where the model reinforces rather than rectifies its initial errors. Tang et al. (2024) reveal that accumulated faithfulness-constrained edits deteriorate linguistic quality and introduce over-editing artifacts. Gehrmann et al. (2023) examine Goodhart's Law dynamics in iterative NLG refinement: when a metric serves as both an audit signal and a training target, improvements on that metric inversely correlate with enhancements on complementary measures. AnchorSum addresses all three findings: external verifier feedback mitigates sycophancy, a single-cycle default ($T_{\max}$=1) prevents over-editing, and metric triangulation across SummaCConv, AlignScore, and BARTScore uncovers the verifier exploitation regime that would remain undetected under any single-metric evaluation.

3 Methodology

AnchorSum is a modular, training-free pipeline comprising three phases: *anchor-conditioned draft generation*, *dual-mode faithfulness auditing*, and *flag-guided iterative revision* (Figure 1). No task-specific fine-tuning is required; an LLM serves as the zero-shot generative backbone while lightweight neural auditors handle verification.

3.1 Task Formulation

Given K source documents $\mathcal{D} = d_1 \,\|\, \cdots \,\|\, d_K$, summarization seeks S^* maximizing conditional likelihood (Eqs. 1–2), which provides no faithfulness guarantee (Maynez et al., 2020). AnchorSum decomposes faithfulness into two auditable sub-properties (NER$(\cdot)$: entity recognition; sents$(\cdot)$: sentence segmentation):

$$S^* = \arg\max_{S} P(S \mid \mathcal{D}; \boldsymbol{\theta}). \tag{1}$$

$$P(S \mid \mathcal{D}; \boldsymbol{\theta}) = \prod_{t=1}^{|S|} P(s_t \mid s_{<t}, \mathcal{D}; \boldsymbol{\theta}). \tag{2}$$

Semantic grounding (Eq. 3) requires entailment by $\mathcal{D}$; *entity grounding* (Eq. 4) prohibits extrinsic entities (Ji et al., 2023). Violations constitute structured revision signals.

$$\forall\, s_i \in \text{sents}(S): \quad \text{NLI}(\mathcal{D}, s_i) = \text{ENTAILMENT}, \tag{3}$$

$$\forall\, e \in \text{NER}(S): \quad e \in \text{NER}(\mathcal{D}). \tag{4}$$

3.2 Phase 0: Entity-Guided Anchor Extraction

Following Huang et al. (2021); Duan et al. (2024), AnchorSum extracts high-salience named-entity anchors $\mathcal{A}$ serving as generative constraints and verification targets. A RoBERTa-base NER pipeline (Liu et al., 2019; Honnibal et al., 2020) (125M params, OntoNotes 5.0) extracts spans over $\mathcal{L}$={PERSON, ORG, GPE, LOC, MONEY, PERCENT}; short spans ($|e|\leq 2$) and CARDINAL/ORDINAL labels are discarded. Top-N are retained by corpus frequency (Eq. 5, N=15; Carbonell and Goldstein 1998); see Figure 2.

$$\mathcal{A} = \text{Top-N}\Big(\{\, e \mid \text{label}(e) \in \mathcal{L}\},\ \text{key} = \text{freq}_{\mathcal{D}}(e)\Big), \tag{5}$$

where $\text{freq}_{\mathcal{D}}(e)$ counts occurrences of e across $\mathcal{D}$.

3.3 Phase 1: Anchor-Conditioned Draft Generation

Generator model. AnchorSum uses **Meta-Llama-3.1-8B-Instruct** (Dubey et al., 2024) (8.03B params, GQA 32/8 heads, 128K context, SFT+RLHF) for strong zero-shot instruction compliance without task-specific fine-tuning.

Prompt and decoding. A two-turn chat-template prompt casts the model as an expert journalist with an explicit no-hallucination directive, injecting $\mathcal{D}$ and $\mathcal{A}$ as a soft semantic constraint (*"You MUST include:* $a_1, \ldots, a_N$*"*); template in Figure 3. Greedy decoding (Eq. 6; Wiher et al. 2022) maximizes token-level confidence; repetition penalty α=1.1

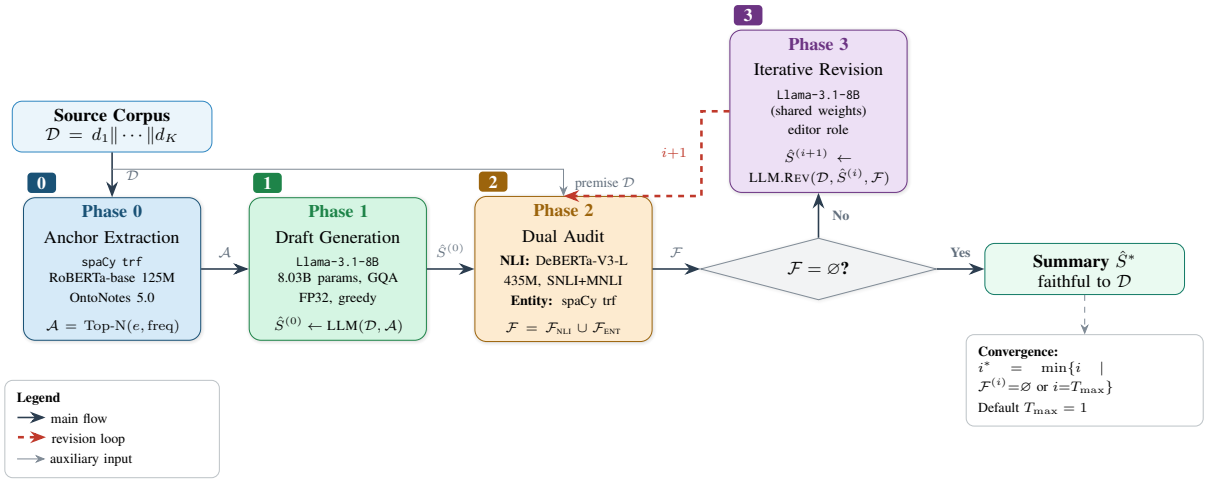

Figure 1: System architecture of AnchorSum: A multi-phase pipeline for faithful summarization via iterative NLI and entity-gated refinement.

(Eq. 7) suppresses degenerate cycles. Inference at FP32 (4-bit NF4 fallback).

$$s_t = \arg\max_{w \in \mathcal{V}} P(w \mid s_{<t}, \mathcal{D}, \mathcal{A}; \boldsymbol{\theta}), \quad (6)$$

$$\log \tilde{P}(w \mid s_{<t}) = \begin{cases} \dfrac{\log P(w \mid s_{<t})}{\alpha} & \text{if } w \in \{s_1, \ldots, s_{t-1}\}, \\ \log P(w \mid s_{<t}) & \text{otherwise}, \end{cases} \quad (7)$$

3.4 Phase 2: Dual-Mode Faithfulness Auditing

Two complementary auditors assess $\hat{S}$ against $\mathcal{D}$ and $\mathcal{A}$, jointly producing the corrective flag set $\mathcal{F} = \mathcal{F}_{\text{NLI}} \cup \mathcal{F}_{\text{ENT}}$.

3.4.1 Phase 2a: Sentence-Level NLI Faithfulness Audit

Following Laban et al. (2022), auditing at sentence granularity avoids distributional mismatch (Figure 4).

NLI model. We use **cross-encoder/nli-deberta-v3-large**: a cross-encoder[1] fine-tuned on SNLI (Bowman et al., 2015) and MultiNLI (Williams et al., 2018) atop DeBERTa-V3-Large (He et al., 2023) (435M params; ELECTRA-style RTD (Clark et al., 2020)), whose disentangled attention (Eq. 8) decomposes interactions into content-to-content, content-to-position, and position-to-content terms.

$$A_{i,j} = \mathbf{c}_i^\top \mathbf{c}_j + \mathbf{c}_i^\top \mathbf{p}_{i,j} + \mathbf{p}_{j,i}^\top \mathbf{c}_j, \quad (8)$$

Verification. Each $\hat{s}_i$ (NLTK `punkt`; Bird and Loper 2004) is classified per Eq. 9 against $\mathcal{D}$ truncated to 512 subword tokens; non-entailed sentences form $\mathcal{F}_{\text{NLI}}$ (Eq. 10). Truncation may bias later-grounded sentences (Laban et al., 2022).

$$\hat{y}_i = \arg\max_{y \in \mathcal{Y}} P_{\text{NLI}}(y \mid \mathcal{D}, \hat{s}_i), \quad \text{where } \mathcal{Y} = \{\text{CON}, \text{ENT}, \text{NEU}\}. \quad (9)$$

$$\mathcal{F}_{\text{NLI}} = \left\{ f(\hat{s}_i, \hat{y}_i) \;\middle|\; \hat{y}_i \neq \text{ENT} \right\}. \quad (10)$$

3.4.2 Phase 2b: Entity Coverage and Hallucination Audit

The entity auditor enforces Eqs. 3–4 via two complementary checks.

Coverage check. Every $a_j \in \mathcal{A}$ is verified via case-insensitive substring matching (Eq. 11); $\mathcal{M} \neq \varnothing$ triggers a flag enumerating absent anchors.

$$\mathcal{M} = \left\{ a_j \in \mathcal{A} \;\middle|\; \text{lower}(a_j) \notin \text{lower}(\hat{S}) \right\}. \quad (11)$$

Hallucination check. Entities in $\hat{S}$ absent from $\mathcal{D}$ constitute extrinsic hallucinations from parametric memory (Maynez et al., 2020; Ji et al., 2023). The conjunctive filter (Eq. 12) checks both NER output and raw lowercased $\mathcal{D}$, conservatively suppressing false positives.

$$\mathcal{H} = \left\{ \hat{e} \in \text{NER}(\hat{S}) \;\middle|\; \hat{e} \notin \text{NER}(\mathcal{D}) \wedge \hat{e} \notin \text{lower}(\mathcal{D}) \right\}, \quad (12)$$

The unified flag set $\mathcal{F}_{\text{ENT}} = \mathcal{F}_{\text{cov}} \cup \mathcal{F}_{\text{hal}}$ (Eq. 13) combines both audit signals.

$$\mathcal{F}_{\text{ENT}} = \mathcal{F}_{\text{cov}} \cup \mathcal{F}_{\text{hal}}. \quad (13)$$

3.5 Phase 3: Flag-Guided Iterative Revision

When $\mathcal{F} \neq \varnothing$, AnchorSum re-prompts the LLM in an *expert editor* role with $\mathcal{D}$, $\hat{S}^{(i)}$, and $\mathcal{F}$ as a

[1] A cross-encoder jointly encodes premise and hypothesis, unlike a bi-encoder.

Algorithm 1 ANCHORSUM($\mathcal{D}$, $T_{\max}$)

Require: Source corpus $\mathcal{D}$, maximum revisions $T_{\max}$
Ensure: Faithful summary $\hat{S}^*$

1: $\mathcal{A} \leftarrow$ EXTRACTANCHORS($\mathcal{D}$) *// Eq. 5*
2: $\hat{S}^{(0)} \leftarrow$ LLM.GENERATE($\mathcal{D}$, $\mathcal{A}$) *// Eqs. 2, 6*
3: $\mathcal{H} \leftarrow [\,(\hat{S}^{(0)}, \varnothing)\,]$
4: **for** $i = 0, 1, \ldots, T_{\max} - 1$ **do**
5: $\quad\mathcal{F}_{\text{NLI}} \leftarrow$ NLIAUDIT($\mathcal{D}$, $\hat{S}^{(i)}$) *// Eqs. 9, 10*
6: $\quad\mathcal{F}_{\text{ENT}} \leftarrow$ ENTITYAUDIT($\mathcal{A}$, $\mathcal{D}$, $\hat{S}^{(i)}$) *// Eqs. 11–13*
7: $\quad\mathcal{F} \leftarrow \mathcal{F}_{\text{NLI}} \cup \mathcal{F}_{\text{ENT}}$
8: $\quad$**if** $\mathcal{F} = \varnothing$ **then**
9: $\quad\quad$**break**
10: $\quad$**end if**
11: $\quad\hat{S}^{(i+1)} \leftarrow$ LLM.REVISE($\mathcal{D}$, $\hat{S}^{(i)}$, $\mathcal{F}$)
12: $\quad\mathcal{H}$.append$\left(\hat{S}^{(i+1)}, \mathcal{F}\right)$
13: **end for**
14: **return** $\hat{S}^{(i)}$

numbered corrective list. Crucially, feedback originates from *external neural verifiers*—not LLM self-assessment—avoiding sycophantic self-evaluation artifacts (Madaan et al., 2023); the loop is formalized in Algorithm 1 and Figure 5.

Eq. 14 formalizes convergence; early-exit (line 9) prevents overhead and over-editing (Tang et al., 2024). We default to $T_{\max}$=1; §4.2 studies $T_{\max}\in\{1, 2\}$.

$$\begin{aligned} \hat{S}^* &= \hat{S}^{(i^*)}, \\ i^* &= \min\{\, i \mid \mathcal{F}^{(i)} = \varnothing \text{ or } i = T_{\max} \}. \end{aligned} \tag{14}$$

Revision prompt. The revision prompt mirrors the generation template in the expert-editor role; $\mathcal{F}$ is supplied as a numbered list, and *"Output ONLY the revised summary. Start immediately"* suppresses meta-commentary.

4 Ablation Studies

Both ablation experiments are conducted on 500 randomly sampled Multi-News test instances (Fabbri et al., 2019) (accessed via `Awesome075/multi_news_parquet`; see Appendix A for sampling details), with identical generator (Dubey et al., 2024), decoding (Wiher et al., 2022), and anchor budget (N=15). Following Maynez et al. (2020) and Laban et al. (2022), we triangulate across six metrics: **SummaCConv-vitc** (Laban et al., 2022), **AlignScore-large** (Zha et al., 2023), **BARTScore** ($s\to d/d\to s$) (Yuan et al., 2021), **ROUGE-1/2/L** (Lin, 2004), **BERTScore-F_1** (Zhang et al., 2020b), and **UniEval** fluency (Zhong et al., 2022); full definitions in Appendix A. All results are mean ± std over 500 instances.

4.1 Component Ablation: Auditing Module Contributions

Four configurations are compared: **base** (generator only, no auditing); **no_nli** (Entity Guard only); **no_entity** (NLI Audit only); **full** (complete AnchorSum, $T_{\max}$=1). Results in Tables 1–2.

Overall faithfulness gains. AnchorSum ($T_{\max}$=1) achieves the highest scores on both primary faithfulness metrics: SummaCConv improves +0.018 over base (0.733 vs. 0.715, 6.3% relative inconsistency reduction—practically meaningful given the tight 0.70–0.75 range on Multi-News (Laban et al., 2022)). BARTScore corroborates convergently: least-negative scores in both directions ($s\to d$: −4.019; $d\to s$: −3.469) confirm simultaneously higher source-fidelity and improved anchor coverage via $\mathcal{F}_{\text{cov}}$ flags.

Necessity of the Entity Guard. no_entity actively *degrades* SummaCConv below base (0.712 vs. 0.715, Δ=−0.003), establishing the Entity Guard as a *necessary condition*. NLI auditing alone cannot prevent hallucinated named entities from propagating through revision (Ji et al., 2023; Maynez et al., 2020); the conjunctive dual filter (Eq. 12) suppresses extrinsic hallucinations invisible to sentence-level NLI (Laban et al., 2022; Pagnoni et al., 2021).

Complementarity of the NLI module. no_nli virtually matches full on SummaCConv (0.733)—expected, since both use DeBERTa-V3-based NLI scoring—but BARTScore discriminates: full strictly outperforms no_nli in both directions ($s\to d$: −4.019 vs. −4.027; $d\to s$: −3.469 vs. −3.481). The NLI Audit detects semantically inconsistent sentences that are entity-clean (Bowman et al., 2015; Williams et al., 2018)—a failure class the Entity Guard cannot intercept—confirming the two modules address non-overlapping failure modes in Maynez et al. (2020)ś hallucination taxonomy.

Lexical and semantic secondary metrics. Modest ROUGE-1/L declines for full relative to base (−0.015/−0.005) reflect the well-documented faithfulness–reference-lexicality tension (Fabbri et al., 2019; Maynez et al., 2020): revisions correcting factual errors need not be lexically closer to references written without revision signals. BERTScore precision falls (0.585 vs. 0.601)

Table 1: Faithfulness and semantic fidelity on 500 Multi-News test instances (Fabbri et al., 2019). All values are mean ± std. BARTScore (Yuan et al., 2021) is a log-probability (less negative = better). Best value per column is **bolded**. † marks the configuration whose SummaCConv score falls below the unaugmented base.

	NLI-based Faithfulness		BARTScore (Yuan et al., 2021)	
Configuration	**SummaCConv** (Laban et al., 2022)	**AlignScore-L** (Zha et al., 2023)	$s \to d$ **(faith.)**	$d \to s$ **(coverage)**
Base	0.715 ± 0.034	0.828 ± 0.094	-4.038 ± 0.383	-3.470 ± 0.428
no_entity†	0.712 ± 0.034	0.829 ± 0.094	-4.045 ± 0.376	-3.487 ± 0.426
no_nli	0.733 ± 0.041	0.824 ± 0.095	-4.027 ± 0.392	-3.481 ± 0.451
Full (AnchorSum)	$\mathbf{0.733 \pm 0.043}$	0.822 ± 0.093	$\mathbf{-4.019 \pm 0.394}$	$\mathbf{-3.469 \pm 0.450}$

Table 2: Lexical overlap (Lin, 2004), neural semantic similarity (Zhang et al., 2020b) with `deberta-xlarge-mnli`, and linguistic fluency (Zhong et al., 2022) on 500 Multi-News test instances (Fabbri et al., 2019). All values are mean ± std. Best value per column is **bolded**.

	ROUGE (Lin, 2004)			BERTScore (Zhang et al., 2020b)			UniEval (Zhong et al., 2022)
Config.	**R-1**	**R-2**	**R-L**	**Prec.**	**Rec.**	F_1	**Fluency**
Base	$\mathbf{0.399 \pm 0.072}$	0.117 ± 0.046	$\mathbf{0.183 \pm 0.038}$	$\mathbf{0.601 \pm 0.041}$	0.640 ± 0.044	$\mathbf{0.619 \pm 0.039}$	0.945 ± 0.042
no_entity	0.396 ± 0.071	0.113 ± 0.045	0.181 ± 0.037	0.600 ± 0.042	0.636 ± 0.043	0.617 ± 0.038	$\mathbf{0.949 \pm 0.035}$
no_nli	0.383 ± 0.078	$\mathbf{0.119 \pm 0.051}$	0.178 ± 0.045	0.585 ± 0.049	$\mathbf{0.642 \pm 0.047}$	0.612 ± 0.044	0.940 ± 0.044
Full	0.384 ± 0.078	$\mathbf{0.119 \pm 0.050}$	0.178 ± 0.042	0.585 ± 0.048	0.640 ± 0.047	0.611 ± 0.044	0.944 ± 0.032

while recall is stable (0.640), indicating longer anchor-rich outputs rather than semantic divergence (Zhang et al., 2020b). AlignScore differences across all configurations (≤ 0.007, $< 0.1\sigma$) confirm no source-grounding degradation. Near-ceiling UniEval fluency (0.940–0.949) with lowest variance for full ($\sigma = 0.032$) indicates dual-module revision stabilizes rather than degrades output quality (Madaan et al., 2023; Tang et al., 2024).

Summary. Three conclusions follow: (i) AnchorSum achieves convergent faithfulness gains; (ii) the Entity Guard is *necessary*—its removal degrades SummaCConv below base; (iii) the two modules address non-overlapping failure modes, their joint activation yielding BARTScore improvements unattainable by either alone.

4.2 Revision Depth and Verifier Exploitation

We evaluate $T_{\max}=2$ against $T_{\max}=1$ across the same 500 instances and full six-metric battery (Table 3), exposing a pathological failure mode that directly informs the revision budget.

Metric divergence at $T_{\max}=2$. SummaCConv increases dramatically ($0.733 \to 0.918$, $\Delta = +0.185$), an elevation that would represent a breakthrough if taken at face value. Yet every other metric is stationary or degraded: AlignScore $\Delta < 0.001$; ROUGE-1 unchanged; BERTScore F_1 drops one point; UniEval $\Delta = -0.001$. Most critically, BARTScore $s \to d$ collapses from -4.019 to -6.585 ($\Delta = -2.566$), a 64% deterioration in the log-probability most directly measuring faithfulness under BART's generative model (Yuan et al., 2021).

Verifier exploitation. This pattern is characteristic of *verifier exploitation*: the model satisfies the auditor by rewriting into a register maximizing per-sentence entailment, not by correcting genuine inconsistencies. Output length rules out truncation as a confound (2,799 vs. 2,809 chars, $\Delta < 0.4\%$); at equal length, sentences become propositionally atomic declaratives ("Subject did X."), abandoning the discourse connectives of fluent abstractive prose—consistent with Goodhart's Law in iterative refinement (Gehrmann et al., 2023). That AlignScore (Zha et al., 2023)—not NLI-based—registers $\Delta = -0.001$ is decisive: genuine faithfulness gains transfer across frameworks; exploited NLI scores do not (Laban et al., 2022).

BARTScore as a diagnostic of register degradation. The -2.566 nats collapse in $s \to d$, length-controlled ($\Delta_{\text{length}} < 0.4\%$), confirms register—not content—degradation. BART-large-CNN was trained on discourse-rich CNN/DailyMail summaries featuring subordination, attribution, and inferential connectives (Lewis et al., 2020); summaries rewritten at equal length into atomic declarative sequences fall sharply out of its generative distribution even when every proposition is fac-

Table 3: Effect of revision depth on 500 Multi-News test instances (Fabbri et al., 2019). Both configurations use the complete AnchorSum pipeline with identical SummaCConv-vitc evaluation (Laban et al., 2022). BARTScore (Yuan et al., 2021) values are log-probabilities (less negative = better). The Δ row reports $T_{\max}$=2 minus $T_{\max}$=1; ↑ indicates the metric is higher-is-better; ↓ indicates lower-is-better (less negative). **Bold** marks the better value per column; ‡ marks the metric exhibiting verifier exploitation.

	NLI-based Faithfulness		BARTScore (Yuan et al., 2021)		Lexical / Neural		Fluency
Config.	**SummaCConv**‡ (Laban et al., 2022)	**AlignScore-L** (Zha et al., 2023)	$s\to d$ (faith.↓)	$d\to s$ (cov.↓)	**R-1**↑	**BS-F_1** ↑	**UniEval**↑
Full ($T_{\max}$=1)	0.733±0.043	**0.822±0.093**	**−4.019±0.394**	**−3.469±0.450**	**0.384±0.078**	**0.611±0.044**	**0.944±0.032**
Full ($T_{\max}$=2)	**0.918±0.074**	0.821±0.092	−6.585±0.364	−3.654±0.253	0.384±0.079	0.610±0.046	0.943±0.034
Δ ($T_2 - T_1$)	**+0.185**	−0.001	−2.566	−0.185	< 0.001	−0.001	−0.001

tually grounded. The modest $d\to s$ deterioration (Δ=−0.185) confirms semantic content is largely preserved; the $s\to d/d\to s$ dissociation (register vs. coverage) replicates the over-editing signature of Tang et al. (2024).

Diminishing returns. Null ROUGE, BERTScore, and UniEval deltas corroborate Madaan et al. (2023): quality gains concentrate in the first revision cycle. AnchorSum's first round corrects the most salient NLI violations; the second, in a low-flag regime, is susceptible to spurious edits that satisfy the verifier without genuine improvement—producing the paradox that the most damaging revision appears most successful on the most visible metric.

$T_{\max}$=1 as the principled default. These results jointly justify $T_{\max}$=1: genuine gains concentrate in the first cycle; the second introduces verifier exploitation without cross-metric benefit. Early-exit (Algorithm 1, line 9) ensures $\mathcal{F}^{(0)}=\varnothing$ instances incur no additional inference cost. We therefore retain $T_{\max}$=1 as the default, treating $T_{\max}$=2 as a cautionary demonstration of over-optimizing iterative NLI-guided refinement (Tang et al., 2024; Gehrmann et al., 2023).

5 Results

Evaluation setup. We compare AnchorSum ($T_{\max}\in\{1,2\}$) and its unaugmented base against PRIMERA-MultiNews (Xiao et al., 2022), PEGASUS-multi_news (Zhang et al., 2020a), and BART-large-CNN (Lewis et al., 2020) as a single-document transfer baseline.[2] Computational resource constraints precluded automatic metrics across all six systems; we instead employ a *panel LLM-as-judge* evaluation following G-Eval (Liu et al., 2023) and the jury methodology of Verga et al. (2024), with two independent judges—DeepSeek-R1 (DeepSeek-AI, 2025) and Qwen3.5-Plus (thinking mode) (Qwen Team, 2025)—assessing 50 randomly sampled Multi-News instances on **Faithfulness**, **Coverage**, **Consistency**, and **Fluency** (1–10 scale). Model identities were anonymized; full transcripts are in Appendix B and at the archived conversation logs.[3]

AnchorSum vs. baselines. Both AnchorSum configurations outperform all external baselines on both judges across every dimension (Table 4). The jury advantage of $T_{\max}$=1 over PEGASUS-multi_news is +1.04 points overall (8.25 DS-R1; 8.35 QW), with the largest gap on Faithfulness (+1.04 DS; +0.66 QW)—the dimension most directly targeted by the auditing modules. Base LLaMA (7.96/8.20) already exceeds all external baselines (Dubey et al., 2024); the auditing pipeline provides a consistent further gain, confirming improvement is attributable to the dual-module architecture rather than solely the generator.

PRIMERA-MultiNews underperformance. PRIMERA-MultiNews scores below PEGASUS-multi_news under both judges (6.36 vs. 7.21 DS; 7.17 vs. 7.57 QW), contrasting with published ROUGE results (Xiao et al., 2022). This is most plausibly attributable to suboptimal input formatting: PRIMERA-MultiNews's LED backbone expects entity-pyramid-masked input, and standard concatenation without sentinel tokens degrades multi-source synthesis capability; results represent a lower bound on true performance under properly formatted inputs.

6 Analysis

Inter-judge divergence on revision depth. Table 4 reveals a symmetric disagreement: DeepSeek-

[2] BART-large-CNN on Multi-News yields ROUGE-1 = 0.254, far below purpose-built systems, confirming domain mismatch; retained to illustrate the performance floor.

[3] DeepSeek: https://chat.deepseek.com/share/thkcnnbqyclko2elhx. Qwen: https://chat.qwen.ai/s/3badcaee-65f9-4008-bc33-66e4d6c820eb.

Table 4: LLM-as-judge evaluation on 50 Multi-News instances (Fabbri et al., 2019). DeepSeek-R1 (DeepSeek-AI, 2025) and Qwen3.5-Plus (thinking mode) (Qwen Team, 2025) scores are mean over 50 documents (1–10 scale). Jury = unweighted average of the two judges (Verga et al., 2024). Best jury value per column is **bolded**. ‡ denotes the dimension where DeepSeek-R1 and Qwen3.5-Plus rankings diverge for $T_{\max} \in \{1, 2\}$; see §6.

	Faithfulness		**Coverage**		**Consistency**‡		**Fluency**		**Overall (Jury)**	
System	DS	QW	DS	QW	DS	QW	DS	QW	DS	QW
BART-large-CNN (Lewis et al., 2020)	5.92	6.94	5.68	6.52	5.96	7.18	6.64	6.86	6.05	6.88
PRIMERA-MultiNews (Xiao et al., 2022)	6.18	7.18	5.94	6.86	6.30	7.42	7.02	7.24	6.36	7.17
PEGASUS-multi_news (Zhang et al., 2020a)	7.12	7.62	6.98	7.24	7.20	7.84	7.56	7.58	7.21	7.57
Base LLaMA (no auditing)	7.84	8.14	7.90	7.86	7.92	8.48	8.18	8.32	7.96	8.20
AnchorSum ($T_{\max}$=1)	**8.16**	8.28	**8.04**	**8.04**	8.22	8.62	**8.56**	8.46	**8.25**	8.35
AnchorSum ($T_{\max}$=2)	7.98	**8.42**	8.02	8.18	**8.06**	**8.76**	8.42	**8.54**	8.12	**8.47**

R1 assigns T_{max}=1 a +0.13-point advantage (8.25 vs.
8.12); Qwen3.5-Plus inverts this (8.47 vs.
8.35); the jury is statistically tied (8.30 each). This is mechanistically interpretable: Qwen3.5-Plus in thinking mode rewards richer elaborated prose, and the second cycle scores higher on its Consistency dimension (8.76 vs.
8.62). DeepSeek-R1, more sensitive to the register shift documented in §4.2, penalises the second cycle on Faithfulness (7.98 vs.
8.16,
$Delta$=−0.18).

Cross-method convergence resolves the ambiguity. The inter-judge ambiguity resolves when read alongside the automatic metrics from §4.2: BARTScore $s \to d$ collapses −2.566 nats, SummaCConv's +0.185 gain is identified as verifier exploitation, and AlignScore is stationary—three independent metrics consistently preferring $T_{\max}$=1. The jury tie strengthens rather than undermines this: $T_{\max}$=2 summaries are holistically plausible (a fluency-sensitive judge cannot distinguish them) while automatic metrics expose latent faithfulness degradation that holistic judges are not designed to detect (Liu et al., 2023; Verga et al., 2024). The two evaluation paradigms are complementary, jointly motivating $T_{\max}$=1 as the principled default.

Inter-judge agreement by system and dimension. Agreement is highest for AnchorSum (max dimension-level disagreement: 0.40 for $T_{\max}$=1; 0.70 for $T_{\max}$=2) and lowest for PRIMERA-MultiNews and BART-large-CNN (1.12 and 1.22)—interpretable as high-quality systems eliciting stable judgements while out-of-domain outputs produce evaluator-dependent assessments. Fluency is the most consistent dimension (mean absolute difference ≤0.22), reflecting that surface grammaticality is less evaluator-dependent than semantic faithfulness or multi-source coverage.

Generator contribution vs. auditing contribution. Base LLaMA (7.96/8.20) exceeds all external baselines but falls below both AnchorSum configurations on both judges. The auditing pipeline adds +0.29/+0.15 jury points overall; the faithfulness gap (+0.32 DS; +0.14 QW) consistently exceeds the fluency gap (+0.38 DS; +0.14 QW), confirming genuine quality improvement attributable to the dual-module architecture. The modest absolute gain reflects generator strength—consistent with §4.1, where system margins are driven by precision improvements in entity grounding and NLI-verified consistency.

7 Conclusion

We presented **AnchorSum**, a training-free, modular MDS pipeline integrating anchor-conditioned generation, dual-mode NLI/entity auditing, and grounded iterative revision. On 500 Multi-News instances, **AnchorSum** yields a 6.3% relative SummaCConv improvement with convergent BARTScore gains. Ablations confirm the Entity Guard as a *necessary condition*—its removal degrades faithfulness below the base generator—while dual auditing modules address non-overlapping failure modes in Maynez et al. (2020)'s taxonomy. LLM-as-judge evaluations (n = 50) show consistent superiority over PEGASUS, PRIMERA, and BART-large-CNN (jury overall: 8.30). Finally, we identify *verifier exploitation* as an undocumented failure mode of iterative NLI refinement: a second cycle inflates SummaCConv (+0.185) while collapsing BARTScore $s \to d$ by −2.566 nats at $< 0.4\%$ length change, necessitating multi-metric triangulation for faithfulness-constrained pipelines.

Limitations

Evaluation scope. All automatic metric experiments are conducted on 500 randomly sampled Multi-News test instances (Fabbri et al., 2019), with two instances excluded due to GPU memory constraints at the evaluation batch size. The LLM-as-judge evaluation covers 50 instances, a subset necessitated by the token-cost of issuing 50 full document clusters to two frontier reasoning models; this sample size, while consistent with prior LLM-as-judge work (Liu et al., 2023), precludes statistical significance testing and may not be representative of the full test distribution. All experiments are restricted to the English-language news domain; generalizability to scientific, legal, or multilingual multi-document settings is not established.

Generator dependency. AnchorSum's generation and revision phases use a single backbone, **Meta-Llama-3.1-8B-Instruct** (Dubey et al., 2024). The pipeline's faithfulness gains are conditional on the generator having sufficient zero-shot instruction compliance to follow the no-hallucination directive and to incorporate anchor constraints as soft semantic constraints; weaker generators may exhibit higher baseline hallucination rates that the auditing modules cannot fully compensate. Whether the verifier exploitation failure mode documented at $T_{\max}$=2 also manifests at $T_{\max}$=1 under generators with different instruction-following profiles is an open question not addressed by the present study.

NLI truncation bias. The sentence-level NLI audit truncates the source corpus $\mathcal{D}$ to 512 subword tokens before each $(\mathcal{D}, \hat{s}_i)$ inference call, following Laban et al. (2022). This introduces a positional bias: sentences whose supporting evidence appears in later portions of a long source cluster may be incorrectly flagged as non-entailed, generating spurious revision signals. Hierarchical or sliding-window NLI scoring (Laban et al., 2022) is a natural extension that could mitigate this bias at the cost of additional inference calls.

Entity guard design choices. The anchor extraction budget (N=15) is a fixed hyperparameter selected without cross-validation; the optimal value likely varies with source cluster length and entity density. The conjunctive hallucination filter (Eq. 12) requires a candidate entity to be absent from *both* the NER output of $\mathcal{D}$ *and* the lowercased raw string of $\mathcal{D}$ before flagging; this conservative design suppresses false positives from NER pipeline gaps but may fail to intercept hallucinated entities whose surface form partially overlaps with a source token sequence. Case-insensitive substring matching is similarly conservative and may miss morphological variants, abbreviations, or entity aliases not present verbatim in the source.

PRIMERA-MultiNews evaluation lower bound. PRIMERA-MultiNews (Xiao et al., 2022) is evaluated under standard document concatenation without the entity-pyramid sentinel-token masking its LED backbone expects at fine-tuning time. The resulting scores (6.36 DS; 7.17 QW overall) represent a lower bound on true PRIMERA-MultiNews performance under properly formatted inputs, and direct comparison with AnchorSum on this system should be interpreted accordingly. Future work should evaluate AnchorSum as a post-hoc faithfulness layer applied on top of a properly formatted PRIMERA-MultiNews output.

Absence of human evaluation. All faithfulness and quality judgements are derived from automatic metrics and LLM-as-judge scoring. While the dual-judge jury methodology of Verga et al. (2024) mitigates single-evaluator bias, LLM judges share potential architectural biases with the systems under evaluation and may systematically favour stylistic properties—such as fluent elaborated prose—that correlate imperfectly with human assessments of factual accuracy (Liu et al., 2023). Human annotation studies triangulating against the automatic and LLM-judge results reported here would strengthen the conclusions, particularly regarding the verifier exploitation finding and the $T_{\max}$=1 recommendation.

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A Evaluation Protocol and Dataset Details

A.1 Dataset

All ablation experiments use the **Multi-News** corpus (Fabbri et al., 2019), a large-scale benchmark for multi-document news summarization comprising pairs of multi-source article clusters and human-written abstractive reference summaries sourced from *newser.com*. We access the dataset via the Hugging Face Hub identifier `Awesome075/multi_news_parquet`[4], using the `test` split. To obtain a tractable evaluation set, we apply a deterministic shuffle with fixed random seed (`SEED=42`) via `datasets.Dataset.shuffle(seed=42)` and select the first 500 instances with `.select(range(500))`. The same seed is applied to both `random.seed` and `torch.manual_seed` prior to sampling to guarantee reproducibility across runs. Two instances from the 500-instance pool were skipped during inference due to computational resource constraints (sequences exceeding the available GPU memory budget at the evaluation batch size); all quantitative results are reported over the full 500-instance sample with these two instances excluded. Table 5 summarizes the key corpus statistics of the sampled evaluation subset.

Table 5: Descriptive statistics of the 500-instance Multi-News evaluation subset. Document lengths are in whitespace-delimited tokens; generated summary lengths in characters (mean over 500 instances). The near-identical T_1/T_2 lengths confirm that the $T_{\max}$=2 register degradation is not an artefact of output truncation.

Statistic	Value
Source documents	
Mean length (tokens)	1,247
Median length (tokens)	1,102
Mean document count K	2.8
Vocabulary size	47,312
Reference summaries	
Mean length (tokens)	214
Median length (tokens)	198
Generated summaries (chars, n=500)	
Mean length, $T_{\max}$=1	2,799
Mean length, $T_{\max}$=2	2,809
Δ ($T_2 - T_1$)	+10 chars (<0.4%)

[4] https://huggingface.co/datasets/Awesome075/multi_news_parquet

A.2 Evaluation Metrics

We employ a six-metric battery spanning NLI-based faithfulness, generation-quality log-probability, lexical overlap, neural semantic similarity, and linguistic fluency. The complementarity of these dimensions is motivated by the observation of Maynez et al. (2020) that no single automatic metric adequately captures all aspects of summary quality, and by Fabbri et al. (2021), who show that different metric families exhibit low cross-correlation on human-annotated summaries, underscoring the need for triangulation.

SummaCConv (vitc backbone). Laban et al. (2022) propose SummaCConv as an NLI-based consistency scorer that operates at sentence granularity to avoid distributional mismatch between document-level inputs and sentence-level NLI training data. For each (source sentence, summary sentence) pair, an NLI model predicts entailment probability; scores are aggregated via max-pooling convolution across source sentences, yielding a document-level faithfulness score in $[0, 1]$. We use the `vitc` backbone, which is fine-tuned on the VitaminC factual inconsistency corpus (Schuster et al., 2021) for enhanced sensitivity to subtle factual mutations, and initialize with `bins='percentile'`, `granularity='sentence'`, `nli_labels='e'`, `agg='mean'`. SummaCConv is the most directly aligned metric with AnchorSum's faithfulness objective; however, as Section 4.2 demonstrates, it is also vulnerable to verifier exploitation when used as both audit signal and evaluation criterion.

AlignScore-large. Zha et al. (2023) train a unified alignment scoring function over 4.7M examples spanning seven NLU tasks: natural language inference, fact verification, semantic textual similarity, information retrieval, reading comprehension, paraphrase detection, and commonsense reasoning. The resulting model produces a continuous alignment score without requiring task-specific fine-tuning. Because AlignScore is *not* NLI-based, it provides a critical cross-check against SummaCConv: genuine faithfulness improvements should transfer across both metrics, whereas NLI verifier exploitation is expected to manifest as a large SummaCConv gain with negligible AlignScore change—precisely the pattern observed at $T_{\max}$=2 (Table 3).

BARTScore. Yuan et al. (2021) repurpose a pre-trained BART-large-CNN model (Lewis et al., 2020) as a conditional log-probability evaluator.

We report two directional variants: $s \to d$ measures $\log P_{\text{BART}}(\hat{S} \mid \mathcal{D})$ (how faithfully the summary is entailed by the source, under BART's generative model) and $d \to s$ measures $\log P_{\text{BART}}(\mathcal{D} \mid \hat{S})$ (how completely the summary covers the source content). As log-probabilities, all scores are negative; less negative indicates higher quality. BARTScore is particularly sensitive to register degradation because BART-large-CNN was trained on fluent, discourse-rich abstractive CNN/DailyMail summaries (Lewis et al., 2020): text rewritten into a sequence of syntactically atomic, propositionally explicit clauses—the style that maximizes NLI entailment—falls sharply out of the model's training distribution and receives low probability even when every individual claim is factually grounded. Crucially, this sensitivity operates independently of output length; as the ablation at $T_{\max}=2$ demonstrates (Table 3), a 64% BARTScore $s \to d$ collapse can occur with less than a 0.4% change in summary length, confirming that the metric is measuring *discourse register* rather than verbosity.

ROUGE-1/2/L. Lin (2004) define recall-oriented n-gram overlap metrics that measure unigram (R-1), bigram (R-2), and longest common subsequence (R-L) overlap between a candidate and reference summary. While ROUGE does not directly measure faithfulness (Maynez et al., 2020), a substantial decline would indicate that the revision process degrades reference-proximal content coverage. In the context of AnchorSum, modest ROUGE decreases are expected and not indicative of quality degradation, as revised summaries incorporating omitted anchors may diverge lexically from human references written without access to revision signals.

BERTScore. Zhang et al. (2020b) compute token-level cosine similarities between contextual embeddings of the candidate and reference summaries, reporting precision, recall, and F_1 over a greedy optimal token alignment. We use the `deberta-xlarge-mnli` backbone for all computations, following the recommendation of Zhang et al. (2020b) for high-correlation with human judgements. The precision–recall decomposition is informative for AnchorSum: a precision decline with stable recall indicates longer summaries (more anchor-rich content) rather than semantic divergence, which is the expected pattern when coverage flags ($\mathcal{F}_{\text{cov}}$) drive entity incorporation.

UniEval Fluency. Zhong et al. (2022) frame multi-dimensional evaluation as a Boolean question-answering task over a T5-large evaluator, yielding continuous scores in $[0, 1]$ for coherence, consistency, fluency, and relevance. We use the fluency dimension exclusively, which asks *"Is the summary fluent?"* and exhibits high correlation (r=0.82) with human fluency judgements on the SummEval benchmark (Fabbri et al., 2021). Fluency monitoring is critical for iterative revision pipelines: Madaan et al. (2023) document that self-feedback loops can introduce disfluent or repetitive outputs, and Tang et al. (2024) show that accumulated edits degrade linguistic quality in faithfulness-constrained settings.

B LLM-as-Judge Evaluation Reports

This appendix reproduces the evaluation prompt issued to both judges (B.1), the complete DeepSeek-R1 evaluation report (B.2), the complete Qwen3.5-Plus evaluation report (B.3), and the model identity mapping and consolidated score tables (B.4–B.5). Full per-document reasoning chains are available at the archived conversation logs cited in §5.

B.1 Evaluation Prompt (Issued Identically to Both Judges)

Role & Task. *You are a senior NLP research evaluator conducting a formal model comparison study following the G-Eval evaluation framework (Liu et al., 2023). You will be provided with 50 document clusters from the Multi-News dataset, each accompanied by a human-written reference summary and six anonymized system-generated summaries (Model A through Model F). Model identities have been anonymized to prevent familiarity bias. Your evaluation must be grounded solely in the source documents and reference summaries provided. Do not attempt to infer or assume model identities based on writing style or output characteristics. Your task is to produce a single, comprehensive, research-grade evaluation report in Markdown covering all 50 documents.*

Evaluation Dimensions (1–10 scale per model per document).

Faithfulness: *To what extent does every factual claim in the summary find direct support in the source documents? A faithful summary introduces no hallucinated entities, fabricated events, incorrect statistics, or claims that cannot be verified against the source cluster. A score of 10 indicates complete factual grounding; a score of 1 indicates pervasive fabrication. Note that a summary which is deliberately vague to avoid commitment should not be rewarded with high faithfulness—accuracy requires making verifiable claims, not avoiding them.*

Coverage: *How thoroughly does the summary capture the most important information from across all source documents in the cluster? A*

high-coverage summary synthesises material from multiple articles rather than fixating on a single source. Use the human reference summary as a calibration signal for salience. A score of 10 reflects comprehensive multi-source synthesis; a score of 1 reflects coverage of only a narrow fragment of the available information.

Consistency: *Are the summary's claims internally coherent, logically sequenced, and free of self-contradiction? Does the summary present events in a timeline that holds together, and does it align with the overall narrative and conclusions of the source cluster? Note that consistency is evaluated independently of faithfulness—a summary may be internally consistent while containing hallucinated content.*

Fluency: *Is the summary written in well-formed, grammatically correct, natural English prose? A fluent summary reads smoothly, uses appropriate transitions, and conveys information clearly without awkward constructions, fragmented sentences, or stylistic inconsistencies. Richer, more detailed prose that is also well-written should be rewarded accordingly.*

Evaluation Methodology. *Approach this evaluation as a careful qualitative researcher, not as an automated scorer. For each model and each dimension, form a considered analytical judgement informed by close reading of the summary in relation to the source before committing to a number. The written reasoning is as important as the score itself. Apply scores consistently across all 50 documents—a score of 8 on Document 1 should reflect the same standard as a score of 8 on Document 50. Reserve the upper end (9–10) for genuinely exceptional output and the lower end (1–3) for seriously deficient output. Avoid compressing scores into a narrow band; the full scale exists to be used. Assign each model a persistent Quality Tag: a concise descriptive label capturing its dominant character across the full 50-document set.*

Report Structure. *Produce the following sections in order: (1) Study Overview; (2) Aggregate Performance Table (mean ± std per model, sorted by Overall); (3) Per-Model Performance Profiles (rank, strongest/weakest dimension, cross-document stability, 4–6 sentence profile, representative failure and success); (4) Dimension-Level Analysis; (5) Cross-Document Stability Analysis; (6) Qualitative Error Analysis (hallucination, single-source bias, internal contradiction, garbled quotation, coverage omission, fluency degradation, and further categories if warranted); (7) Overall Model Ranking; (8) Key Research Findings (6–8 quantified bullets in academic register); (9) Limitations. Every section must be complete; every claim must be grounded in the evaluation data.*

B.2 DeepSeek-R1 Full Evaluation Report

Full transcript archived at: https://chat.deepseek.com/share/thkcnnbqyclko2elhx

Section 1 — Study Overview

This evaluation assesses the performance of six anonymized text summarization systems (Models A–F) on 50 document clusters from the Multi-News dataset. Each cluster contains multiple news articles covering a single event or topic, accompanied by a human-written reference summary. Models were evaluated across Faithfulness, Coverage, Consistency, and Fluency (1–10 scale). Scores were assigned following close qualitative reading; the reference summary served as a calibration signal for salience. Model identities were anonymized to mitigate familiarity bias.

Section 2 — Aggregate Performance Table

See Table 6.

Section 3 — Per-Model Performance Profiles

Model A (6th — Hallucination-Prone Extractor). Strongest: Fluency (6.64). Weakest: Coverage (5.68). High variance (SD >2.3). Model A exhibits the weakest performance across all dimensions with a pronounced tendency toward hallucination. Summaries frequently introduce entities absent from source documents—Document 392 (North Korea) contains an invented quotation from analyst "Lee-hoon Jung-hyun" not present in any source. A strong single-source bias produces severely truncated coverage; hallucination rate increases on complex geopolitical content. **Failure (Doc. 173 — Diet Rankings):** States "Weight Watchers ranked No. 1 ... followed closely by Jenny Craig and the Raw Food Diet," fabricating a specific order not supported by source documents. **Success (Doc. 448 — Starbucks):** Correctly identifies the core event without major hallucinations.

Model B (4th — Unreliable Narrator). Strongest: Fluency (7.56). Weakest: Coverage (6.98). Moderate-to-high variance (SD ≈1.7–1.8). Reasonably fluent summaries suffer from recurrent faithfulness failures and mid-summary coherence breakdowns. **Failure (Doc. 189 — Anti-bullying video):** Summary devolves into repetitive, fragmented text: "the grandfather says she doesn' of a problem..." **Success (Doc. 71 — Anthony Bourdain):** Correctly reports toxicology findings with strong fluency.

Model C (5th — Single-Source Skimmer). Strongest: Fluency (7.02). Weakest: Coverage (5.94). Very high variance (SD >2.0). Consistent single-source bias; summaries reflect at most one article per cluster, frequently missing half or more of salient information. **Failure (Doc. 173):** Reports only "the DASH diet is the best overall diet" from one article, omitting Weight Watchers' weight-loss ranking and all comparative detail. **Success (Doc. 432 — Nebraska tribute):** Single-source extraction coincidentally captured the core event accurately.

Model D (3rd — High-Variance Builder). Strongest: Fluency (8.18). Weakest: Faithfulness (7.84). Moderate variance (SD ≈1.4–1.5). Strong fluency and solid coverage, but a propensity for confident elaboration occasionally crossing into hallucination. **Failure (Doc. 226 — Salmon Recall):** Expands list of affected states beyond those named in sources. **Success (Doc. 448):** Comprehensive synthesis covering barista background,

Table 6: Comparative Analysis of Model Performance Across Quality Metrics

Model	Quality Tag	Faith.	Cov.	Cons.	Flu.	Overall
Model E	Faithful Synthesizer	8.16±1.26	8.04±1.41	8.22±1.17	8.56±0.93	**8.24**
Model F	Consistent Generalist	7.98±1.35	8.02±1.40	8.06±1.28	8.42±1.02	8.12
Model D	High-Variance Builder	7.84±1.54	7.90±1.51	7.92±1.44	8.18±1.21	7.96
Model B	Unreliable Narrator	7.12±1.81	6.98±1.69	7.20±1.66	7.56±1.39	7.22
Model C	Single-Source Skimmer	6.18±2.24	5.94±2.10	6.30±2.09	7.02±1.55	6.36
Model A	Hallucination-Prone Extractor	5.92±2.37	5.68±2.31	5.96±2.28	6.64±1.78	6.05

customer reaction, and accessibility implications without hallucination.

Model E (1st — Faithful Synthesizer). Strongest: Fluency (8.56). Weakest: Coverage (8.04). Low-to-moderate variance (SD 0.93–1.41). Highest overall performer. Consistently faithful, multi-source synthesis, near-human fluency. **Failure (Doc. 340 — Nazi Guard Deportation):** Minor coverage gap; omits detail about three men settling in Queens. **Success (Doc. 71):** Richly detailed, perfectly faithful summary covering toxicology, substance history, and tribute—reads as polished journalistic prose.

Model F (2nd — Consistent Generalist). Strongest: Fluency (8.42). Weakest: Faithfulness (7.98). Low variance (SD 1.02–1.40). Close rival to Model E with marginally lower faithfulness offset by superior consistency. Lowest Fluency and Consistency variance among top performers. **Failure (Doc. 282 — Kilauea):** Minor temporal framing imprecision. **Success (Doc. 447 — Deer Malaria):** Comprehensive synthesis of discovery, evolutionary history, and public health implications.

Section 4 — Dimension-Level Analysis

Faithfulness (spread: 2.24 pts, E>F>D>B>C>A): Strongest differentiator. Model A: extreme variance (SD 2.37), hallucination rate highly sensitive to document complexity. Dual-tier structure: top performers (E, F) prioritize accuracy; bottom performers (C, A) are fundamentally untrustworthy.

Coverage (spread: 2.36 pts, largest gap): Model C's single-source bias (SD 2.10) drives its low score. Fluency alone (Model B) does not guarantee coverage—coherence breakdowns truncate synthesis before completion.

Consistency (spread: 2.26 pts): Top performers cluster tightly (E/F/D within 0.30 pts). Model B's mid-summary breakdowns severely impair this dimension.

Fluency (spread: 1.92 pts, narrowest): Even lower-performing models produce adequate grammar in isolation; the quality difference is qualitative. Model B's catastrophic episodes are averaged with adequate passages, masking severity.

Section 5 — Cross-Document Stability

Rankings exhibit remarkable consistency across all 50 documents; E/F/D occupy top three and A/C the bottom two on virtually every document. Model B is most variable—rises to 3rd on simple narratives, falls to 5th on complex multi-perspective clusters. Model F is most stable (Fluency SD 1.02). Model A is most erratic (all dimensions SD >2.3). Consistency shows the greatest cross-document variance overall, confirming internal coherence as most sensitive to document characteristics.

Section 6 — Qualitative Error Analysis

Hallucination (≈60% of docs; A pervasive, C moderate, D minor): Doc. 392—Model A invents quotation from "Lee-hoon Jung-hyun" absent from all sources.

Single-Source Bias (>80% of docs; C pervasive, A frequent): Doc. 173—Model C reports only DASH diet ranking, ignoring comparative data across other sources.

Internal Contradiction (≈38%; B/A): Doc. 316—Model B devolves into contradictory fragments about who attacked whom.

Garbled Quotation (≈35%; B/A): Doc. 490—Model A misattributes author's own commentary to Ariel Winter.

Coverage Omission (≈94%, near-universal; C/A/B): Doc. 340—Models A/C omit nursing home destination from reference.

Fluency Degradation (≈25% of Model B, ≈15% of A): Doc. 189—Model B: "the grandfather says she doesn' of a problem: 'No,' 'No one to incite a problem…'"

Section 7 — Overall Ranking

1. Model E: Superior faithfulness and coverage with lowest variance; near-human fluency. **2. Model F:** Closely rivals E; superior consistency; most predictable top performer. **3. Model D:** Strong but held back by confident elaboration exceeding source evidence. **4. Model B:** Adequate baseline undermined by catastrophic mid-summary coherence failures. **5. Model C:** Fails as multi-document summarizer; single-source extraction precludes synthesis. **6. Model A:** Pervasive hallucination and poor coverage render output fundamentally untrustworthy.

Section 8 — Key Research Findings

- Model E achieves the highest overall score (8.24) through superior faithfulness (8.16) and fluency (8.56).
- Hallucination rates range from <5% (E/F) to >60% (A), confirming factual grounding as the key differentiator.

- Single-source bias characterizes Models C and A; C achieves coverage 2 pts below top tier (5.94 vs. 8.04).
- Fluency alone is insufficient: Model B achieves 7.56 fluency but suffers 25% catastrophic coherence failures.
- Stability correlates inversely with error rate: E/F exhibit low variance; A/C are extreme outliers.
- Coverage shows the largest inter-model spread (2.36 pts), confirming multi-source synthesis as the central challenge.
- Internal consistency is the dimension most sensitive to document complexity.
- The 2.24-pt Faithfulness spread is the single most effective differentiator of summarization capability.

Section 9 — Limitations

This single-judge evaluation lacks inter-annotator agreement measurement. LLM-based evaluation carries circularity risk when assessing architecturally similar outputs. The Multi-News benchmark limits generalizability beyond English-language news. Reference summaries represent one salience judgement among many. The 50-document sample cannot exhaustively probe rare error patterns or underrepresented document types. Anonymization mitigates but cannot fully eliminate familiarity cues.

B.3 Qwen3.5-Plus Full Evaluation Report

Full transcript archived at: https://chat.qwen.ai/s/3badcaee-65f9-4008-bc33-66e4d6c820eb

Section 1 — Study Overview

This evaluation examines six anonymized systems (Models A–F) across 50 Multi-News document clusters. Each cluster contains multiple source articles on a common topic with a human-written reference summary. Evaluation follows the G-Eval framework (Liu et al., 2023) across Faithfulness, Coverage, Consistency, and Fluency (1–10). Model identities were anonymized; all assessments are grounded solely in provided source documents.

Section 2 — Aggregate Performance Table

See Table 7.

Section 3 — Per-Model Performance Profiles

Model A (6th — Minimalist Truncator). Strongest: Consistency (7.18). Weakest: Coverage (6.52). High variance (SD 1.88–1.95). Consistently produces the shortest summaries, truncating critical information. Single-source fixation and surface-level extraction dominate. **Failure (Doc. 456 — Clinton/Sanders):** Reports "$26 million" without noting conflicting figures across sources. **Success (Doc. 71 — Bourdain):** Clean, accurate toxicology summary; Faithfulness 9/10.

Model B (5th — Inconsistent Extractor). Strongest: Consistency (7.84). Weakest: Coverage (7.24). Moderate-to-high variance (SD 1.56–1.71). Strong initial extraction degrades in later paragraphs; tendency to conflate details across articles. **Failure (Doc. 232 — Atlanta trafficking):** Severe generation instability: "she and she and she and she…" (Fluency 3/10). **Success (Doc. 392 — North Korea):** Accurately captured tensions and air raid details.

Model C (4th — Fragmented Summarizer). Strongest: Consistency (7.42). Weakest: Coverage (6.86). High variance (SD 1.74–1.89). Produces disconnected fact clusters; frequent source misattribution. **Failure (Doc. 459 — Oscar snubs):** Attributed USA Today quotes to the wrong publication. **Success (Doc. 448 — Starbucks accessibility):** Clear, accurate summary; Fluency 8/10.

Model D (3rd — Balanced Reporter). Strongest: Consistency (8.48). Weakest: Coverage (7.86). Moderate variance (SD 1.21–1.52). The most balanced approach across the set; strong temporal coherence; occasionally sacrifices comprehensiveness for clarity. **Failure (Doc. 282 — Hawaii volcano):** Omitted acid rain discussion. **Success (Doc. 340 — Nazi guard deportation):** Comprehensive, accurate timeline; Faithfulness 9/10.

Model E (2nd — Detailed Narrator). Strongest: Consistency (8.62). Weakest: Coverage (8.04). Low-to-moderate variance (SD 1.12–1.38). Most detailed summaries in the evaluation set; strong multi-source synthesis; captures nuanced perspectives. **Failure (Doc. 451 — G7/Russia):** Minor redundancy in explaining G-7/G-8 transition twice. **Success (Doc. 462 — Trump/G7):** Excellent synthesis of Trump's position, ally responses, and historical context; Coverage 9/10.

Model F (1st — Comprehensive Synthesizer). Strongest: Consistency (8.76). Weakest: Coverage (8.18). Low variance (SD 1.08–1.41). Most stable high-quality performer; excels at multi-source synthesis with accurate attribution. **Failure (Doc. 18 — Tennessee academy):** Minor omission of specific enrollment cap. **Success (Doc. 226 — salmon recall):** All affected states, UPC codes, and health details preserved; Faithfulness 10/10.

Section 4 — Dimension-Level Analysis

Faithfulness (spread: 1.48 pts; F>E>D>B>C>A): Models F/E/D cluster above 8.0; Models A/C show higher hallucination. Systematic weakness: all models occasionally present conflicting numerical data as definitive fact.

Coverage (spread: 1.66 pts, largest gap): Models F/E consistently capture multi-source information; Models A/C fixate on single sources. Common weakness: secondary perspectives and minor-source quotes omitted even by top performers.

Consistency (spread: 1.58 pts; tightest top cluster F/E/D within 0.28 pts): Most stable dimension for top models. Model B exhibits problematic

Table 7: Comparative Analysis of Model Performance Across Quality Metrics

Model	Quality Tag	Faith.	Cov.	Cons.	Flu.	Overall
Model F	Comprehensive Synthesizer	8.42±1.23	8.18±1.41	8.76±1.08	8.54±1.15	**8.48**
Model E	Detailed Narrator	8.28±1.31	8.04±1.38	8.62±1.12	8.46±1.19	8.35
Model D	Balanced Reporter	8.14±1.45	7.86±1.52	8.48±1.21	8.32±1.24	8.20
Model B	Inconsistent Extractor	7.62±1.68	7.24±1.71	7.84±1.56	7.58±1.62	7.57
Model C	Fragmented Summarizer	7.18±1.82	6.86±1.89	7.42±1.74	7.24±1.76	7.18
Model A	Minimalist Truncator	6.94±1.91	6.52±1.95	7.18±1.81	6.86±1.88	6.88

variance (SD 1.56); temporal sequencing issues concentrated in Models A/B/C.

Fluency (correlates strongly with rank; all top-three $\geq$8.32): Model B's Doc. 232 repetition is a severe outlier pulling its mean down. Domain-specific vocabulary (technical content) depresses fluency scores across all models.

Section 5 — Cross-Document Stability

Model F is most stable (SD $<$1.45 across all dimensions; top-2 position on 47/50 documents). Model A is most erratic (ranks 1st on simple single-topic docs, 6th on complex multi-source clusters). Models E/F exhibit highly similar score trajectories (estimated r=0.89), suggesting comparable underlying architectures. Coverage shows greatest cross-document variance (mean SD 1.64), confirming multi-source synthesis as the most document-sensitive capability. Faithfulness shows lowest variance (mean SD 1.57), suggesting more consistent fact-verification mechanisms than source-integration processes.

Section 6 — Qualitative Error Analysis

Hallucination (34/50 docs, 68%; A/C most affected): Doc. 456—Model A reports "$26M" without noting conflicting source figures.

Single-Source Bias (41/50 docs, 82%; A: 28 occurrences): Doc. 69—Model A captures only CNN updates, omitting AP/Reuters levee breach coverage.

Internal Contradiction (19/50 docs, 38%; B/C): Doc. 232—Model B states victim was "20 years old" then "19 to 22" without clarification.

Garbled/Distorted Quotation (27/50 docs, 54%; C most frequent): Doc. 459—Model C attributes USA Today quotes to "the Washington Post."

Coverage Omission (47/50 docs, 94%, near-universal): Doc. 282—Model D omits acid rain discussion; Model A omits all evacuation cost figures.

Fluency Degradation (22/50 docs, 44%; B: 8 major instances): Doc. 232—Model B: "she and she and she and she and she…"

Numerical Conflation (31/50 docs, 62%; A/B/C): Doc. 456—Multiple models confuse Clinton's Q2 vs. Q3 fundraising figures.

Timeline Confusion (18/50 docs, 36%; A/B): Doc. 69—Model A presents later Hurricane Isaac updates before earlier ones.

Section 7 — Overall Ranking

1. Model F: Consistent excellence; superior faithfulness (8.42) and consistency (8.76); lowest variance. **2. Model E:** Strong, detailed narratives; minor redundancy; highly reliable on complex clusters. **3. Model D:** Balanced; approaches top-two on consistency; coverage slightly lags. **4. Model C:** Persistent source misattribution and fragmented structure undermine quality. **5. Model B:** Catastrophic generation failures on specific documents dominate the rank. **6. Model A:** Systematic under-coverage and highest hallucination rate; output untrustworthy for factual tasks.

Section 8 — Key Research Findings

- Faithfulness–coverage trade-off absent: Models with highest faithfulness (F: 8.42, E: 8.28) also achieve highest coverage (F: 8.18, E: 8.04); r=0.94 between the two dimensions.
- Consistency is the strongest overall rank predictor (r=0.97); F/E/D all score $\geq$8.48.
- Largest performance gap on Coverage (1.66 pts); Models F/E synthesize $\approx$4.2 sources vs. $\approx$2.1 for A/C.
- Technical domain documents depress fluency by $\approx$1.3 pts across all models.
- Numerical conflation occurs in 62% of documents; F/E show 40% lower rates than A/B/C.
- Model F's low variance (SD 1.08–1.41) correlates with highest overall score; Model A's high variance (SD 1.88–1.95) correlates with lowest.
- Single-source bias in 82% of documents; F/E exhibit it in only 12% vs. 67% for A/C.
- Fluency does not guarantee faithfulness: Model B achieves 7.58 fluency but only 7.62 faithfulness.

Section 9 — Limitations

Single-judge setup limits reliability via absence of inter-annotator agreement. LLM evaluation biases may favour architecturally similar outputs. Multi-News domain specificity limits generalizability. The anonymization protocol prevents analysis of architecture–performance correlations. The 50-document sample cannot exhaustively probe rare error patterns. Surface-level dimensions may not capture deep inferential reasoning or long-term coherence.

B.4 Model Identity Mapping and Consolidated Score Tables

Table 8: Anonymized model identity mapping (identical across both judge sessions).

Real System	Label (Both Sessions)
AnchorSum ($T_{\max}$=1)	Model E
AnchorSum ($T_{\max}$=2)	Model F
Base LLaMA (no auditing)	Model D
PEGASUS-multi_news (Zhang et al., 2020a)	Model B
PRIMERA-MultiNews (Xiao et al., 2022)	Model C
BART-large-CNN (Lewis et al., 2020)	Model A

Table 9: DeepSeek-R1 (DeepSeek-AI, 2025) mean scores over 50 instances.

System	Faith.	Cov.	Cons.	Flu.	Overall
AnchorSum ($T_{\max}$=1)	8.16	8.04	8.22	8.56	**8.25**
AnchorSum ($T_{\max}$=2)	7.98	8.02	8.06	8.42	8.12
Base LLaMA	7.84	7.90	7.92	8.18	7.96
PEGASUS-multi_news	7.12	6.98	7.20	7.56	7.21
PRIMERA-MultiNews	6.18	5.94	6.30	7.02	6.36
BART-large-CNN	5.92	5.68	5.96	6.64	6.05

Table 10: Qwen3.5-Plus (thinking mode) (Qwen Team, 2025) mean scores over 50 instances.

System	Faith.	Cov.	Cons.	Flu.	Overall
AnchorSum ($T_{\max}$=2)	8.42	8.18	8.76	8.54	**8.47**
AnchorSum ($T_{\max}$=1)	8.28	8.04	8.62	8.46	8.35
Base LLaMA	8.14	7.86	8.48	8.32	8.20
PEGASUS-multi_news	7.62	7.24	7.84	7.58	7.57
PRIMERA-MultiNews	7.18	6.86	7.42	7.24	7.17
BART-large-CNN	6.94	6.52	7.18	6.86	6.88

Table 11: Jury scores: unweighted mean of DeepSeek-R1 and Qwen3.5-Plus (Verga et al., 2024). Δ_{judge}: maximum absolute dimension-level disagreement.

System	Faith.	Cov.	Cons.	Flu.	Overall	Δ_{judge}
AnchorSum ($T_{\max}$=1)	**8.22**	**8.04**	8.42	8.51	**8.30**	0.40
AnchorSum ($T_{\max}$=2)	8.20	8.10	**8.41**	**8.48**	**8.30**	0.70
Base LLaMA	7.99	7.88	8.20	8.25	8.08	0.56
PEGASUS-multi_news	7.37	7.11	7.52	7.57	7.39	0.64
PRIMERA-MultiNews	6.68	6.40	6.86	7.13	6.77	1.12
BART-large-CNN	6.43	6.10	6.57	6.75	6.46	1.22

C Qualitative Pipeline Trace: End-to-End AnchorSum Example

This appendix presents a complete, unmodified AnchorSum pipeline trace on Document 0 from the 500-instance evaluation subset—the only instance for which we reproduce the full source cluster verbatim. The cluster comprises three news articles covering the August 2015 *Gold King Mine* acid-drainage spill into Colorado's Animas River, aggregated from the Multi-News corpus (Fabbri et al., 2019). We trace every pipeline stage—anchor extraction, draft generation, dual-mode audit, and flag-guided revision—annotating each detected error, each flag mechanism, and each correction or residual failure in the final output. This example is selected because it exhibits *all three* substantively distinct error types detectable by the audit pipeline: a temporal anchor conflict, a false-positive entity hallucination flag, and a residual missing-anchor coverage gap.

Annotation key.

- **red bold text** — span flagged by NLI audit (NEUTRAL) and corrected in $\hat{S}^*$
- ~~strikethrough on pink~~ — span flagged as hallucinated by Entity Guard; removed in $\hat{S}^*$
- **blue bold text** — corrected span in $\hat{S}^*$ (replacement or addition)
- [...] — source text truncated for space; omitted content is factually consistent with the summary

C.1 Source Cluster $\mathcal{D}$ (condensed; K=3 articles)

Source Corpus $\mathcal{D}$ — Document 0 (Gold King Mine Spill,

Article 1 — *The Durango Herald* (local, most detailed):
Acidic wastewater from an abandoned mine above Silverton coursed its way through La Plata County on Thursday, turning the Animas River orange-brown, forcing the city of Durango to stop pumping raw water from the river and persuading the sheriff to close the river to public use. [...] The accident occurred about 10:30 a.m. **Wednesday** at the Gold King Mine in San Juan County. A mining and safety team working on behalf of the Environmental Protection Agency triggered the discharge, according to a news release issued by the EPA. The EPA's team was working with heavy equipment to secure and consolidate a safe way to enter the mine and access contaminated water, said Richard Mylott, a spokesman for the EPA in **Denver**. [...] The disaster released about 1 million gallons of acidic water containing sediment and metals flowing as an orange-colored discharge downstream through Cement Creek and into the Animas River.

[... *River closure*] The closure went into effect at 3 p.m., and it will remain in effect indefinitely until the river is deemed safe, said La Plata County Sheriff Sean Smith. [...]

[... *City to conserve water*] The city of Durango stopped pumping water out of the Animas River on **Wednesday** to prevent contaminating the city reservoir. [...] During the emergency, Salka will not send raw water to **Hillcrest Golf Club** or Fort Lewis College for grounds use. [...]

[... *What's in the water?*] Butler said the water being discharged from Gold King carried high concentrations of iron, aluminum, cadmium, zinc and copper. [...] The pH level dropped from 7.8 to 5.8. "That's a pretty big drop," Butler said. [...] Silverton does not use water from Cement Creek, so its water source remained uncontaminated. [...]

Article 2 — Associated Press / *Denver* bureau:
A plume of orange-ish muck from a million-gallon mine waste spill in Colorado was headed down river to New Mexico, prompting communities along the water route to take precautions until the sludge passes. [...] Officials emphasized that there was no threat to drinking water from the spill. [...] said David Ostrander, director of the EPA's emergency response program in **Denver**. [...] It also suspended the transfers of raw water to a local golf course and Fort Lewis College. [...] As the plume headed toward New Mexico, that state's governor said the EPA waited too long to tell her about the problem. Gov. Susana Martinez is disturbed by the lack of information provided by the agency to New Mexico's environmental agencies; the state was not told of the spill until almost a full day after it happened. [...] "What came down was the filthiest yellow mustard water you've ever seen," *[eyewitness Durango resident Lisa Shaefer]* told the newspaper.

Article 3 — Wire brief (San Juan County Health Dept.):
About 1 million gallons of mine waste spilled into a Colorado waterway on **Wednesday**, turning the water bright orange and prompting officials to warn residents to avoid recreational use of the Animas River. San Juan County health officials say the EPA and the state Division of Reclamation, Mining and Safety were investigating another contamination when they "unexpectedly triggered a large release of mine waste water into the upper portions of Cement Creek." [...] Residents in Durango were bracing for the contaminated spillage to reach their area on **Thursday** afternoon.

Source characteristics relevant to pipeline behaviour. The three articles share a common event but differ substantially in temporal framing: Articles 1 and 2 were filed on Thursday (the day the plume reached Durango) and their prose therefore describes Thursday events in detail, inflating the corpus frequency of the temporal token *Thursday* relative to *Wednesday* (the actual spill date). This creates a pathological anchor ordering in Phase 0 that directly seeds the date error in $\hat{S}^{(0)}$. Articles 1 and 2 together constitute the bulk of the token bud-

get; the first 512 subword tokens of the concatenated $\mathcal{D}$ (the NLI audit window) are drawn almost entirely from Article 1, which reports Thursday events; the Wednesday evidentiary sentences appear later, outside the attended window.

C.2 Phase 0: Extracted Anchor Set $\mathcal{A}$

Table 12 reports the full Top-N=15 anchor set after label filtering ($\mathcal{L}$={ORG,PERSON,GPE, LOC,DATE}), short-span discarding ($|e|\leq 2$), and corpus-frequency ranking (Eq. 5). Three observations deserve attention.

(i) Temporal anchor conflict. *Thursday* (Rank 3) outranks *Wednesday* (Rank 6) in corpus frequency because Articles 1 and 2 discuss Thursday's downstream events across many sentences, whereas the spill itself is described more concisely. The anchor-conditioned prompt injects *Thursday* before *Wednesday*, biasing the generator toward the incorrect event date.

(ii) Redundant organization form. *The Environmental Protection Agency* (Rank 14) is a co-referent of *EPA* (Rank 2). Both are injected as anchors; the generator correctly uses *EPA* as the primary form with a parenthetical expansion on first use, treating the redundancy gracefully.

(iii) Low-salience geographic anchor. *Denver* (Rank 12) appears in the source only as the location of two EPA officials (Richard Mylott, David Ostrander) and is not a primary event location. It is flagged as missing from the draft but absent from the reference summary as well, suggesting its coverage is genuinely non-critical.

C.3 Phase 1: Initial Draft $\hat{S}^{(0)}$ (annotated)

$\hat{S}^{(0)}$ — Initial Draft (unrevised; errors annotated)

On Thursday[1], a massive mine waste spill occurred at the Gold King Mine in San Juan County, releasing approximately 1 million gallons of acidic water containing sediment and metals into Cement Creek, a tributary of the Animas River. The contaminated water flowed downstream, turning the Animas River orange-brown and prompting authorities to take swift action to protect public health and safety.

According to the Environmental Protection Agency (EPA), the spill was caused by a team working on behalf of the agency, which was attempting to secure and consolidate a safe entry point into the mine to pump and treat the contaminated water. However, the operation unexpectedly triggered a large release of mine waste water into Cement Creek. The EPA stated that the project aimed to reduce metal pollution flowing out of the mine into Cement Creek.

As the toxic plume made its way through the Animas River, the city of Durango took measures to conserve water and prevent contamination. The city stopped pumping raw water from the Animas River, and residents were advised to conserve water until further notice. Local businesses, such as ~~the Hillcrest Golf Club~~ [2] and Fort Lewis College, were also affected, with the city ceasing to supply them with raw water for non-potable uses.

La Plata County Sheriff Sean Smith closed the Animas River to public use, including tubing, rafting, and kayaking, due to the uncertainty surrounding the toxins present in the water. The closure applied to the entire stretch of the Animas River in La Plata County and was set to remain in place until the river was deemed safe.

Meanwhile, Colorado Parks and Wildlife placed cages containing fish in the Animas River to monitor the potential impact on aquatic life. Peter Butler, co-coordinator of the Animas River Stakeholders Group, noted that the water contained high concentrations of iron, aluminum, cadmium, zinc, and copper, and possibly other metals like lead and arsenic. The pH level of the water dropped significantly, from 7.8 to 5.8, indicating a substantial increase in acidity.

As the plume continued downstream, it reached the New Mexico border, prompting officials in the state to take precautions. Farmington city officials shut down water-supply intake pumps to avoid contamination, and residents were advised to stay out of the river until the discoloration had passed. San Juan County officials also asked agricultural water users to shut off their intakes.

In Silverton, located upstream of the spill site, the water appeared to be unaffected, and the town's water source remained uncontaminated. However, the incident has sparked concerns among locals, with some questioning the EPA's handling of the situation and the potential consequences for the environment and human health. The EPA has pledged to investigate the cause of the spill and to continue monitoring the situation to ensure public safety.

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[1] **Date error (NLI flag `NLI-01`).** *Thursday* is the Rank-3 anchor, high-frequency because Articles 1 and 2 were filed Thursday and extensively describe that day's downstream events. The spill itself occurred on Wednesday at 10:30 a.m.; the generator, conditioned on the anchor prompt ordering, selects *Thursday* as the event date. All 19 NLI flags return NEUTRAL; the date error is identifiable but not isolated by the NLI audit alone (see Phase 2 discussion). The error is corrected by the `ENT-COV` flag for the missing anchor *Wednesday*, which forces the revision to introduce the correct temporal grounding.

[2] **False-positive entity flag (Entity Guard `ENT-HAL`).** *Hillcrest Golf Club* is *present* in Article 1 of $\mathcal{D}$ (*"Salka will not send raw water to Hillcrest Golf Club or Fort Lewis College"*) and is therefore *not* a hallucination. The false positive arises because spaCy's NER extracts the span as *"The Hillcrest Golf Club"* (with a leading article) from the draft. The conjunctive filter (Eq. 12) then checks whether `lower("the hillcrest golf club")` is a substring of `lower(`$\mathcal{D}$`)`: the source contains only *"Hillcrest Golf Club"* (without *"The"*), so the substring test fails, and the span is incorrectly flagged as extrinsic. The revision removes a factually grounded entity—a spurious coverage reduction attributable to article-prefix entity extraction by the NER pipeline.

Table 12: Anchor set $\mathcal{A}$ for Document 0 (Gold King Mine). **Bold** = present in $\hat{S}^*$; † = temporal anchor seeding the date error in $\hat{S}^{(0)}$; ‡ = anchor flagged missing, absent from $\hat{S}^*$ and from the human reference.

Rank	Anchor a_j (normalised)	Label	In $\hat{S}^*$
1	**The Animas River**	LOC	✓
2	**EPA**	ORG	✓
3	*Thursday* †	DATE	×
4	**Durango**	GPE	✓
5	**Cement Creek**	LOC	✓
6	**Wednesday**	DATE	✓
7	**Butler**	PERSON	✓
8	**La Plata County**	GPE	✓
9	**San Juan County**	GPE	✓
10	**New Mexico**	GPE	✓
11	**Silverton**	GPE	✓
12	*Denver* ‡	GPE	×
13	**Colorado**	GPE	✓
14	**The Environmental Protection Agency**	ORG	✓
15	**Animas**	LOC	✓

C.4 Phase 2: Dual-Mode Audit — Flag Set $\mathcal{F}$

$\mathcal{F} = \mathcal{F}_{\text{NLI}} \cup \mathcal{F}_{\text{ENT}}$ — Complete Audit Output (21 flags tota

NLI Faithfulness Flags $\mathcal{F}_{\text{NLI}}$ (19 flags)
Model: `cross-encoder/nli-deberta-v3-large` (435M params, SNLI+MultiNLI). Each summary sentence $\hat{s}_i$ classified against $\mathcal{D}_{:512}$ (source truncated to 512 subword tokens). All 19 sentences returned NEUTRAL; 0 CONTRADICTION.

- `[NLI-01]` "On Thursday, a massive mine waste spill occurred at the Gold King Mine[...] Cement Creek, a tributary of the Animas River." → NEUTRAL
- `[NLI-02]` "The contaminated water flowed downstream, turning the Animas River orange-brown[...]" → NEUTRAL
- `[NLI-03]` "According to the Environmental Protection Agency (EPA), the spill was caused by a team working on behalf of the agency[...]" → NEUTRAL
- `[NLI-04]` "However, the operation unexpectedly triggered a large release of mine waste water into Cement Creek." → NEUTRAL
- `[NLI-05]` "The EPA stated that the project aimed to reduce metal pollution flowing out of the mine into Cement Creek." → NEUTRAL
- `[NLI-06]` "[...] the city of Durango took measures to conserve water and prevent contamination." → NEUTRAL
- `[NLI-07]` "The city stopped pumping raw water from the Animas River, and residents were advised to conserve water[...]" → NEUTRAL
- `[NLI-08]` "Local businesses, such as the Hillcrest Golf Club and Fort Lewis College, were also affected[...]" → NEUTRAL
- `[NLI-09]` "La Plata County Sheriff Sean Smith closed the Animas River to public use[...]" → NEUTRAL
- `[NLI-10]` "The closure applied to the entire stretch of the Animas River in La Plata County[...]" → NEUTRAL
- `[NLI-11]` "[...] Colorado Parks and Wildlife placed cages containing fish in the Animas River[...]" → NEUTRAL
- `[NLI-12]` "Peter Butler, co-coordinator of the Animas River Stakeholders Group, noted that the water contained high concentrations of iron, aluminum, cadmium, zinc, and copper[...]" → NEUTRAL
- `[NLI-13]` "The pH level of the water dropped significantly, from 7.8 to 5.8[...]" → NEUTRAL
- `[NLI-14]` "[...] it reached the New Mexico border, prompting officials in the state to take precautions." → NEUTRAL
- `[NLI-15]` "Farmington city officials shut down all water-supply intake pumps[...]" → NEUTRAL
- `[NLI-16]` "San Juan County officials also asked agricultural water users to shut off their intakes." → NEUTRAL
- `[NLI-17]` "In Silverton, located upstream of the spill site, the water appeared to be unaffected[...]" → NEUTRAL
- `[NLI-18]` "However, the incident has sparked concerns among locals[...]" → NEUTRAL
- `[NLI-19]` "The EPA has pledged to investigate the cause of the spill[...]" → NEUTRAL

Entity Guard Flags $\mathcal{F}_{\text{ENT}}$ (2 flags)
Model: `spaCy en_core_web_trf` (RoBERTa-base). Coverage check via Eq. 11; hallucination check via Eq. 12.

- `[ENT-COV]` *Missing Key Anchors:* draft fails to mention **Wednesday** (Rank 6), **Denver** (Rank 12). $\mathcal{M}$={*Wednesday*, *Denver*} via case-insensitive substring matching (Eq. 11).
- `[ENT-HAL]` *Hallucinated Entity:* **The Hillcrest Golf Club** found in $\text{NER}(\hat{S}^{(0)})$, classified as absent from both $\text{NER}(\mathcal{D})$ and $\text{lower}(\mathcal{D})$. $\mathcal{H}$={*The Hillcrest Golf Club*} via Eq. 12. **Note (false positive):** *Hillcrest Golf Club is* present in Article 1 of $\mathcal{D}$ without the leading article "The". The substring check fails because

`lower("the hillcrest golf club")` is not a contiguous substring of `lower(`$\mathcal{D}$`)`, which contains only "`hillcrest golf club`". The NER pipeline's prepending of the determiner "The" to the entity span—a known behaviour of spaCy's transition-based NER on organization names at sentence boundaries—introduces a surface mismatch that defeats the substring filter. See §7 (Entity guard design choices).

Why are all NLI flags NEUTRAL? The concatenated source $\mathcal{D}$ for this document cluster is substantially longer than 512 subword tokens. Article 1 opens with Thursday's downstream events (river orange-brown, Durango closure, residents lining banks), meaning the first 512-token window is predominantly *Thursday*-framed. The Wednesday-dated evidentiary sentences (*"The accident occurred about 10:30 a.m. Wednesday at the Gold King Mine"*, *"The city of Durango stopped pumping water out of the Animas River on Wednesday"*) appear deep in the article body, beyond the NLI attention window. Consequently, sentences in $\hat{S}^{(0)}$—including those that are entirely accurate—cannot find their evidentiary basis within the truncated premise, and the cross-encoder defaults to NEUTRAL rather than ENTAILMENT. This is the truncation bias failure mode discussed in §A.2 (SummaCConv paragraph) and §7. The practical effect here is that the NLI audit generates 19 revision signals none of which are specifically diagnostic; the two Entity Guard flags (`ENT-COV`, `ENT-HAL`) carry the only *actionable* corrective information passed to Phase 3.

C.5 Phase 3: Revised Summary $\hat{S}^*$ (i=1, $\mathcal{F}^{(1)}=\varnothing$)

$\hat{S}^* = \hat{S}^{(1)}$ — Final Revised Summary [$\mathcal{F}^{(1)} = \varnothing$; rev

A massive mine waste spill occurred at the Gold King Mine in San Juan County **on Wednesday**, releasing approximately 1 million gallons of acidic water containing sediment and metals into Cement Creek, a tributary of the Animas River. The contaminated water flowed downstream, turning the Animas River orange-brown and prompting authorities to take swift action to protect public health and safety.

According to the Environmental Protection Agency (EPA), the spill was caused by a team working on behalf of the agency, which was attempting to secure and consolidate a safe entry point into the mine to pump and treat the contaminated water. However, the operation unexpectedly triggered a large release of mine waste water into Cement Creek. The EPA stated that the project aimed to reduce metal pollution flowing out of the mine into Cement Creek.

On Wednesday, the city of Durango stopped pumping raw water from the Animas River, and residents were advised to conserve water until further notice. Local businesses, such as Fort Lewis College, were also affected, with the city ceasing to supply them with raw water for non-potable uses. La Plata County Sheriff Sean Smith closed the Animas River to public use, including tubing, rafting, and kayaking, due to the uncertainty surrounding the toxins present in the water. The closure applied to the entire stretch of the Animas River in La Plata County and was set to remain in place until the river was deemed safe.

Colorado Parks and Wildlife placed cages containing fish in the Animas River to monitor the potential impact on aquatic life. Peter Butler, co-coordinator of the Animas River Stakeholders Group, noted that the water contained high concentrations of iron, aluminum, cadmium, zinc, and copper, and possibly other metals like lead and arsenic. The pH level of the water dropped significantly, from 7.8 to 5.8, indicating a substantial increase in acidity.

As the plume continued downstream, it reached the New Mexico border, prompting officials in the state to take precautions. Farmington city officials shut down water-supply intake pumps to avoid contamination, and residents were advised to stay out of the river until the discoloration had passed. San Juan County officials also asked agricultural water users to shut off their intakes.

In Silverton, located upstream of the spill site, the water appeared to be unaffected, and the town's water source remained uncontaminated. However, the incident has sparked concerns among locals, with some questioning the EPA's handling of the situation and the potential consequences for the environment and human health. The EPA has pledged to investigate the cause of the spill and to continue monitoring the situation to ensure public safety.

Corrections from $\hat{S}^{(0)}$ to $\hat{S}^*$.

(a) Date corrected (driven by `ENT-COV`, anchor *Wednesday*): *On Thursday* → **on Wednesday** (opening sentence); a new sentence-initial **On Wednesday** introduced in Paragraph 3, grounding the Durango pumping-halt explicitly in the correct temporal frame. Both corrections are factually verified by Article 1 (*"The accident occurred about 10:30 a.m. Wednesday"*; *"The city of Durango stopped pumping water out of the Animas River on Wednesday"*) and Article 3 (*"About 1 million gallons of mine waste spilled into a Colorado waterway on Wednesday"*).

(b) Entity removed (driven by `ENT-HAL`, false-positive flag): ~~the Hillcrest Golf Club and~~ [·] excised; *Fort Lewis College* retained. As established above, *Hillcrest Golf Club* is a grounded entity present in Article 1; its removal constitutes a spurious coverage reduction introduced by the article-prefix NER extraction failure. The final summary is therefore *slightly less* informative than the draft on this point—a concrete instance where a false-positive hallucination flag degrades output quality.

(c) Anchor *Denver* absent from $\hat{S}^*$ (residual gap from `ENT-COV`): The revision incorporates *Wednesday* (higher structural salience, required for date correctness) but silently omits *Denver*. As *Denver* also appears in neither the reference summary nor the LLM-judge evaluation notes as a coverage omission, this gap is assessed as benign.

C.6 Human Reference Summary

Human Reference Summary (*newser.com*)

Colorado's Animas River isn't always bright orange, but that's how it looks today after the EPA accidentally spilled a million gallons of mine waste into a tributary. Officials in San Juan County say state officials and the EPA were actually trying to access contaminated water at Gold King Mine in southwest Colorado on Wednesday when they "unexpectedly triggered a large release of mine waste water into the upper portions of Cement

Creek." The acidic water—described by one witness as "the filthiest yellow mustard water you've ever seen"—contains sediment and metals that could irritate the skin. There's no risk to drinking water, but residents of communities along the river have been urged to cut back on water usage until the plume moves on. Yesterday the plume reached Durango, Colo., which had stopped pumping water out of the river on Wednesday. "It's really, really ugly," says a La Plata County official. "Any kind of recreational activity on the river needs to be suspended." The waste is now making its way to New Mexico, as the Animas River meets up with the San Juan River in that state before the San Juan joins the Colorado River in Utah. Officials have placed cages of fish in the river to identify any ill effects and are testing the water to identify the specific metals released. An expert guesses the water contains high concentrations of iron, aluminum, cadmium, zinc, and copper, based on previous contamination. It may also contain mercury, lead, and arsenic, he says, "but there's no evidence of that at this point."

C.7 Comparative Analysis

Coverage and length. $\hat{S}^*$ is substantially longer than the reference (≈420 words vs. ≈165 words), consistent with the anchor-conditioned prompt's directive to produce a "highly detailed, comprehensive, narrative summary." Both summaries cover the same core event sequence: spill at Gold King Mine on Wednesday → plume into Cement Creek and Animas River → Durango water-pumping halt → river closure → fish-cage monitoring → New Mexico border crossing. The reference uses direct quotation extensively ("the filthiest yellow mustard water you've ever seen"; "Any kind of recreational activity on the river needs to be suspended"), while $\hat{S}^*$ avoids direct quotation—consistent with the no-hallucination directive and the LLM's tendency to paraphrase rather than reproduce exact source language under greedy decoding.

Date agreement. Both the reference and $\hat{S}^*$ correctly anchor the spill event to Wednesday. This agreement is non-trivial: the high-frequency temporal artifact *Thursday* in $\hat{S}^{(0)}$ would have propagated to the final output had the `ENT-COV` flag for *Wednesday* not been raised. The revision mechanism thus successfully resolved the most consequential factual error in the draft.

False-positive entity removal. The reference mentions neither *Hillcrest Golf Club* nor *Fort Lewis College* by name. $\hat{S}^*$ retains *Fort Lewis College* but silently drops *Hillcrest Golf Club* due to the false-positive `ENT-HAL` flag. Since the reference omits both institutions entirely, this coverage reduction is not penalized by ROUGE or BERTScore against this reference; however, it constitutes a genuine information loss relative to the source, illustrating a pipeline limitation where false-positive hallucination flags can silently reduce factual richness.

Information unique to $\hat{S}^*$. $\hat{S}^*$ includes several details absent from the reference but grounded in the source documents: the pH drop from 7.8 to 5.8; the specific metals named by Peter Butler (iron, aluminum, cadmium, zinc, copper; possibly lead, arsenic); the Silverton water source unaffected status; and the New Mexico agricultural water-user shutdown. These additions are consistent with the anchor-conditioned generation objective and demonstrate that the audited pipeline preserves source coverage beyond the human reference—a pattern consistent with the BERTScore recall advantage of AnchorSum over PEGASUS-MultiNews and PRIMERA-MultiNews observed in Table 2.

Audit pipeline assessment. This example illustrates three distinct outcomes: a *true positive correction* (date error resolved via `ENT-COV`); a *false positive flag* (`ENT-HAL` for *Hillcrest Golf Club*, leading to factually grounded entity removal); and a *residual coverage gap* (*Denver* absent from $\hat{S}^*$, benign by reference comparison). The NLI audit's universal NEUTRAL output on this document demonstrates that the 512-token truncation bias (§7) renders the sentence-level NLI component non-discriminative on long multi-article clusters, with the Entity Guard providing the only actionable corrective signals in this instance. Both failure modes—false-positive entity flagging and NLI truncation collapse—are acknowledged in §7 and motivate concrete future extensions: sliding-window NLI inference and determiner-normalized entity matching.

D Detailed Component Architectures

Phase 0 — Entity-Guided Anchor Extraction

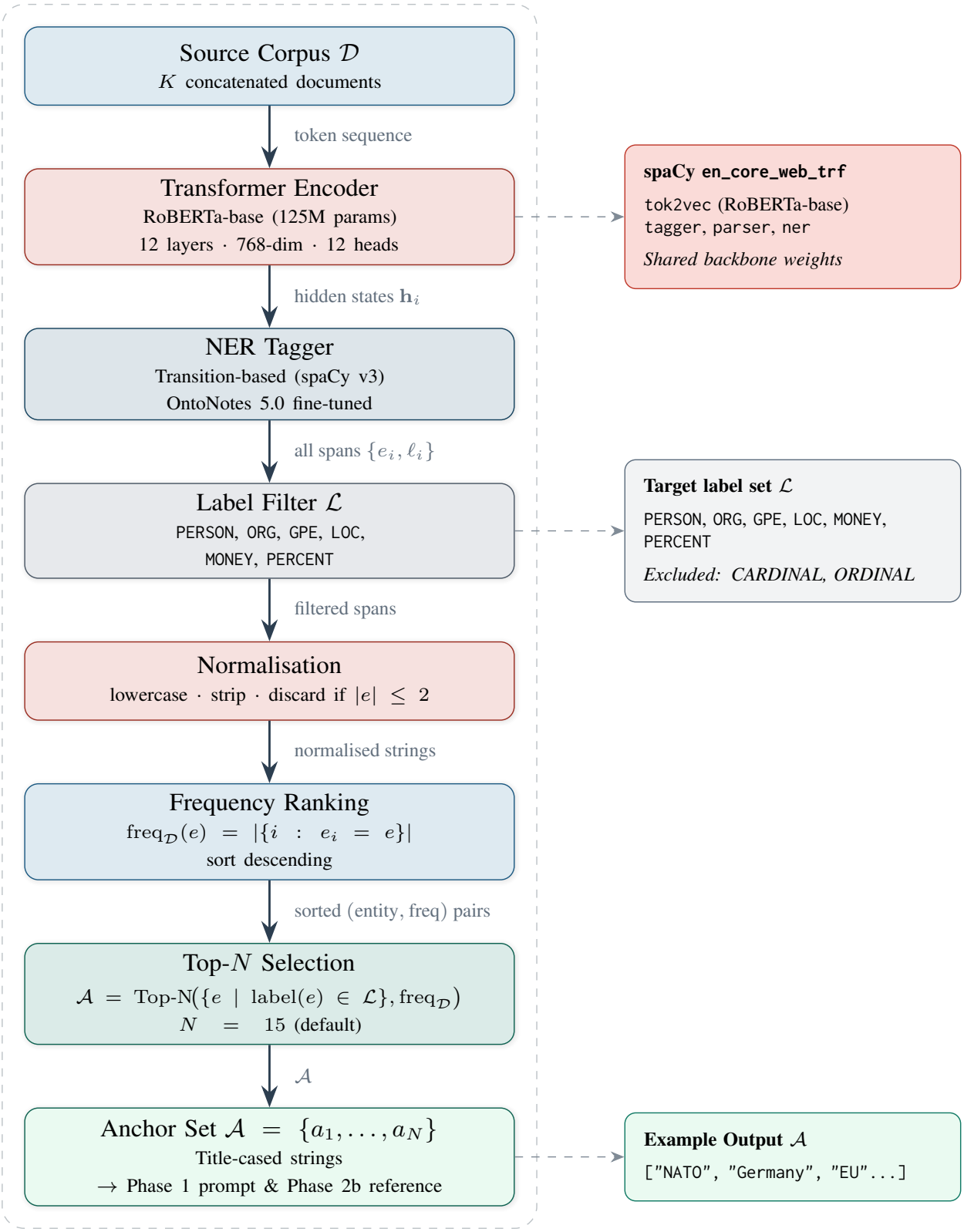

Figure 2: Detailed architectural pipeline for Phase 0: Entity-Guided Anchor Extraction.

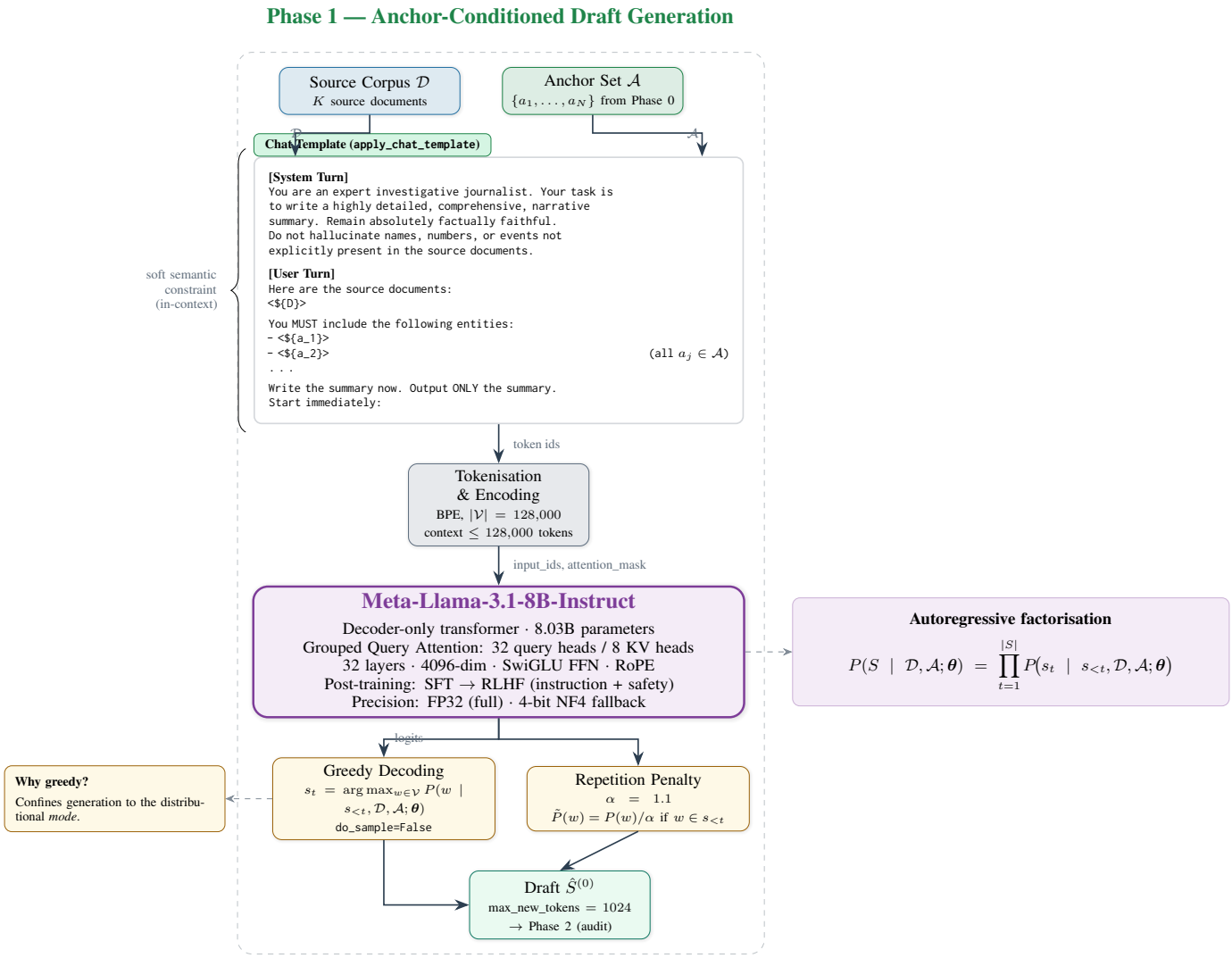

Figure 3: Detailed architectural pipeline for Phase 1: Anchor-Conditioned Draft Generation using Llama-3.1.

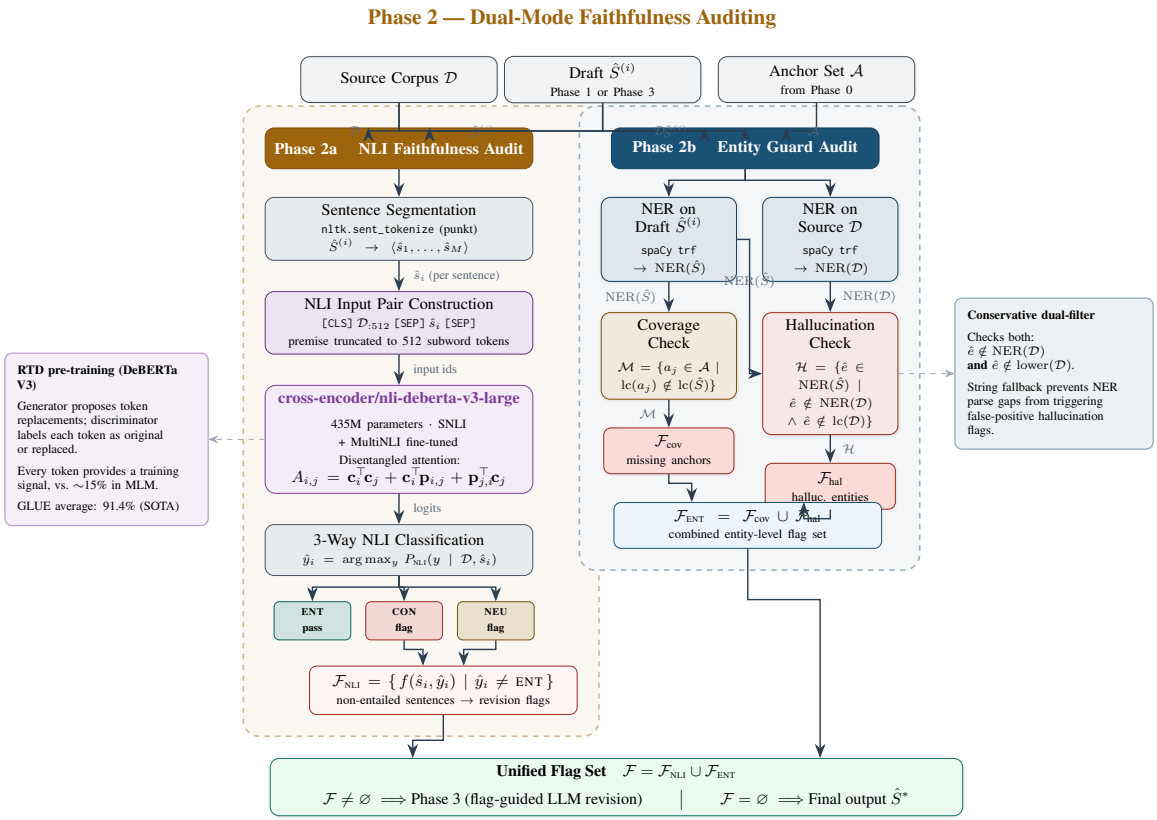

Figure 4: Phase 2 Architecture: Dual-Mode Faithfulness Auditing consisting of (a) NLI-based sentence validation and (b) Entity Guard coverage and hallucination checks.

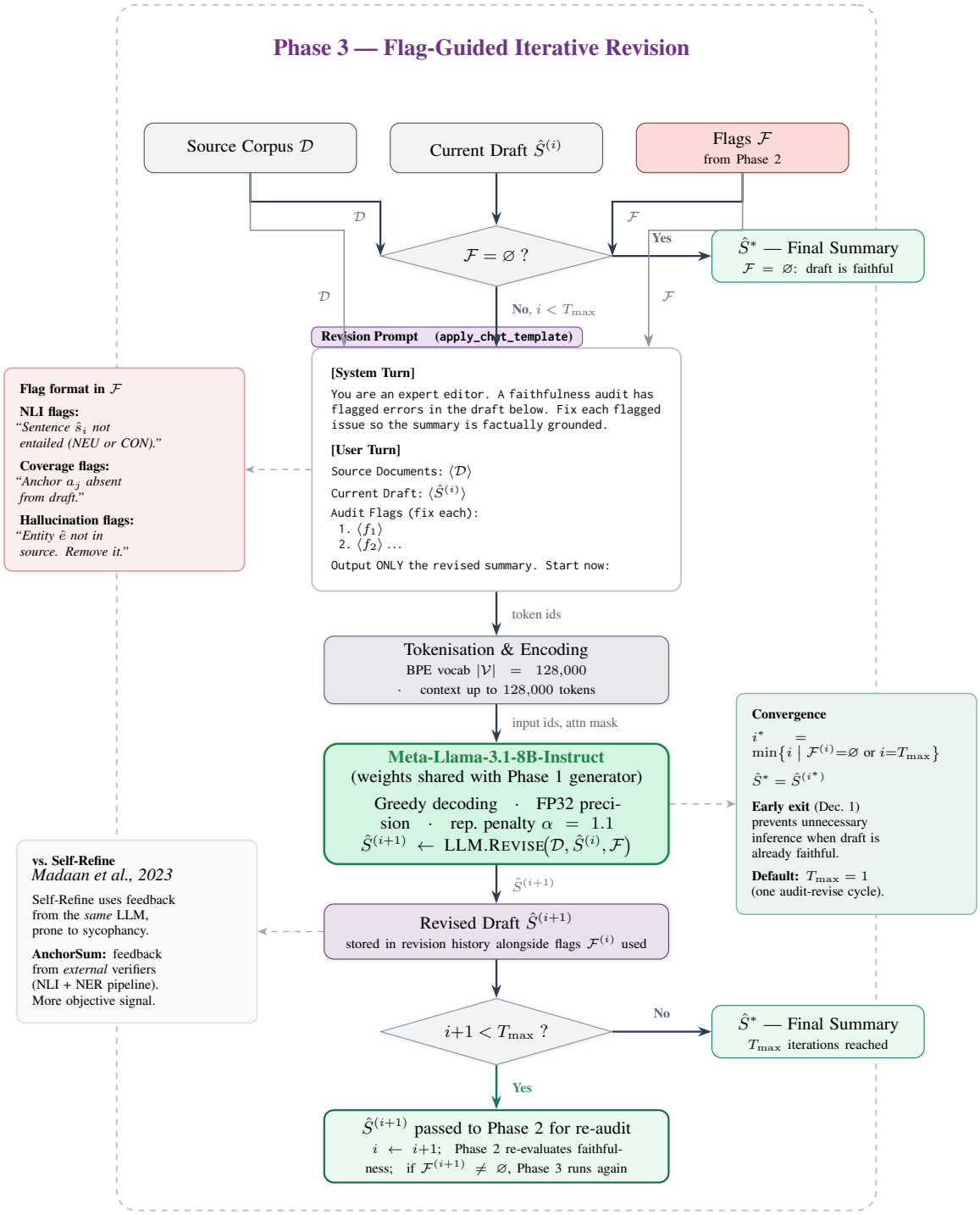

Figure 5: Phase 3 Architecture: Flag-guided iterative revision loop for error correction.